

River Deep Mountain AI

Model Benchmarking Report: Water Quality Risk Mapping (BB2C)

Model Benchmarking Report - Water Quality Risk Mapping (BB2C)

FINAL REPORT

Client: River Deep Mountain AI

Report Date: January 2026

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Document History

Version number	Purpose	Issued by	Quality Checks Approved by	Date
V1.0	Draft issued for comments	Gosia Druzkowska	Amy Jones	12/12/2025
V1.1	Final draft issued for comments	Amy Jones	Daniel Pearce	30/01/2026
V2.0	Final report	Daniel Pearce	Gosia Druzkowska	20/02/2026

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Glossary

BOD	Biochemical Oxygen Demand
CORINE	Coordination of Information on the Environment
CSO	Combined Sewer Overflow
EA	Environment Agency
HAWRAT	Highways Agency Water Risk Assessment Tool
NRW	Natural Resources Wales
RDMAI	River Deep Mountain AI
RNAGs	Reasons for Not Achieving Good Status
RRMSE	Relative Root Mean Square Error
SAAR	Standard Annual Average Rainfall
SAGIS	Source Apportionment-GIS
SEPA	Scottish Environmental Protection Agency
SIMCAT	SIMulation of CATchments
UKWIR	UK Water Industry Research
WFD	Water Framework Directive
WIMS	The EA water quality samples database
WwTW	Wastewater Treatment Work



Summary

Table 1.1 Benchmarking summary

Model Outputs	SAGIS	RDMAI Model
Model Approach	SAGIS (Source Apportionment Geographical Information Systems) provides predictions of river water quality based on pollution source databases. The river quality model SIMCAT is used to predict annual flow and water quality statistics as annual means and percentiles in river catchments by including estimates of river flow and chemical input distributions to river reaches for a set of 38 pollutants. The SIMCAT predictions are derived on a stochastic basis using a combined distributions Monte-Carlo approach. The statistical results can be compared to water quality standards and reported as a compliance assessment.	The RDMAI Risk Map is a geospatial analysis model with some deterministic components. The model ‘feature engineers’ the inputs of various pollutant sources into specific reaches of a waterbody. Examples of the geospatial analyses are determining the closest point along a river reach to a point source and the delineation of upstream catchment for non-point sources. An example of the deterministic component is calculating how pollutants degrade as they move downstream.
Applications	Assessing impact of effluent permit changes/applications on water quality. Source apportionment for water company planning, providing information on background water quality, reasons for failure of water quality standards for each river catchment, and available headroom to inform investment plans. Results relevant	To identify areas within a catchment that are at higher risk of poor river water quality. The risk maps can inform optimal locations for more in-depth investigations and monitoring.



Model Outputs	SAGIS	RDMAI Model
	to this comparison are long-term orthophosphate-P predictions as mean values with confidence intervals.	
Geographic Constraints	SAGIS is built around UK-specific datasets and therefore is suitable for studies in England, Wales and Scotland.	Restricted to England at this stage.
Temporal Resolution	SAGIS provides annual average concentrations for 38 pollutants including phosphate. Results include source apportionment, confidence limits around the mean and percentiles.	Annual average estimate of orthophosphate-P and ammoniacal-N risk.
Data Inputs Required	The source apportionment data and model structures are already set up for the user. There are instructions on editing the model structure and inputs as required. This involves setting up summary statistics (e.g. mean phosphate for wastewater treatment works) or seasonal distributions (e.g. monthly diffuse phosphate loads from agriculture). The data updates are limited to periodic updates coordinated around the 5-year water company investment cycle and the EA requirements to use SAGIS results for the 6-year River Basin Planning and reporting cycle.	The EA consents database needs to be manually downloaded into a stipulated folder. All other datasets are processed automatically by the provided script. Rivers: Open River Network Overflow spill data: EA Storm Overflows - Annual Returns Consents data: EA Consents Database Land use: CROME 2023 dataset Hydrology of Soil Types (HOST) derived from this study.



Model Outputs	SAGIS	RDMAI Model
	The user cannot update all the load estimates as these require inputs from other water quality models and processing other datasets.	Rainfall data: UK Met Office Rainfall Data Roads and buildings: OS OpenMap - Local Built-up areas: OS Open Built Up Areas Farmscoper Data: ADAS Farmscoper Tool is used for agricultural and livestock runoff estimates via lookup tables.
Model Performance	The UKWIR SAGIS model produces relatively representative concentrations of orthophosphate-P for some catchments when compared to observations but does perform badly for other catchments. For all the sites in the benchmark catchments considered, when compared to mean EA measured orthophosphate-P, the UKWIR SAGIS model obtained a relative root mean squared error (RRMSE) of 271%.	The RDMAI Risk Map produces relatively representative concentrations of orthophosphate-P for some catchments when compared to observations but does perform badly for other catchments. For all the sites in the benchmark catchments considered, when compared to mean EA measured orthophosphate-P, the RDMAI Risk Map obtained a relative root mean squared error (RRMSE) of 219%.
Cost & Time	The SIMCAT software and specific catchment model files set up by the EA are free to use. These files would include the source apportionment inputs set up by the EA to represent diffuse and point loads. The SAGIS platform can only be used by experienced water quality modellers and requires access to ArcGIS Pro and purchase of the SAGIS tools. ArcGIS Pro costs around £4,200/yr	The RDMAI Risk Map is open and free to use, as are the input data. The model is virtually 'plug-and-play', so is relatively easy to set up and run. Running a catchment, depending on the size, can take between 5 and 30 minutes on a standard laptop.



Model Outputs	SAGIS	RDMAI Model
	per named licence. A past WRc project allocated around 4 days for using SAGIS to derive water quality baselines. Using an existing SAGIS dataset, such as the UKWIR SAGIS dataset used in this benchmarking, requires a simple download. Extracting the required data from a pre-existing SAGIS dataset takes around 0.5 days.	
Strengths	• Provides predictions of long-term water quality for a wide range of pollutants across all Water Framework Directive (WFD) river reaches for the UK that can be easily compared to statistical water quality standards. • Includes estimates of source apportionment for a comprehensive range of sectors to allow management decisions to be made at catchment and river basin scale. • Continues to be used routinely for quality discharge permit applications to protect rivers from pressures across sectors.	• The model is free and open to use and adapt, and the input data is open and free to use. • Running the model for a catchment is relatively simple and quick. • The river reaches represented in the RDMAI Open Risk Map are based on Ordnance Survey Open Rivers, whereas SAGIS uses the comparatively simpler WFD river reaches. Therefore, the RDMAI Open Risk Map can provide risk for a larger number of river reaches and may be suitable for use by local river action groups. • The benchmarking has shown that the RDMAI Risk Map is at least as accurate at estimating orthophosphate-P as SAGIS for the 12 catchments evaluated. It should be remembered that the result might have been different if other or additional



Model Outputs	SAGIS	RDMAI Model
		catchments were assessed. There is considerable potential to further improve the accuracy of the RDMAI Risk Map by bringing in improved estimates of flow.
Limitations	• The diffuse load inputs cannot easily be updated by users. Relies on periodic updates coordinated across UKWIR, EA and other regulators (NRW/SEPA). This time-lag is a weakness and highlights SAGIS results are appropriate (as designed) for estimates of long-term water quality at the timescale of 5-year water company planning cycle and the 6-year river basin planning cycle. • High cost and skill requirements to set up and use the SAGIS system.	• The approximation of river flow, needed to estimate in-stream pollutant dilution, is rudimentary. This can be resolved by bringing in improved estimates of flow, such as the RDMAI Open Flow Model or QUBE flows. • Estimation of loads from point sources, namely WwTWs and CSOs, is based on open data, primarily the consents database. This leads to a large degree of uncertainty, which can be reduced by incorporating water company data. • At present, the geographical coverage of the model is limited to England. Due to the differences in data availability between the devolved nations, expansion of coverage would necessitate changes to the methodology. • Risk maps can only be created for orthophosphate-P and ammoniacal-N.





1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Risk Map Overview

The RDMAI Risk Map methodology aims to provide a standardised and open approach to map catchment risks, to identify areas within a catchment that are at higher risk of poor river water quality. The risk maps can inform optimal locations for more in-depth investigations and monitoring.

The model represents certain key 'pressures' affecting water quality in England's rivers, informed by the Environment Agency's (EAs) Reasons for Not Achieving Good status (RNAGs).

These are:

- Sewage discharge (continuous), i.e., Wastewater Treatment Works (WwTWs) final effluent discharges.
- Sewage discharge (intermittent), i.e., storm overflow spills.
- Agricultural runoff.
- Urban and road runoff – considering rural and urban roads within 200 m of a river reach (trunk roads) and runoff from urban areas.
- Private Sewage Treatment.

The risk maps also represent a limited number (two) of water quality variables, informed again by the RNAGs:

- Reactive Orthophosphate as P (hereafter, orthophosphate-P).
- Ammoniacal Nitrogen as N (the sum of NH_4 and NH_3) (hereafter, ammoniacal-N).

In calculating risk in a catchment, the model follows several steps, the most important being:

1. Separating the rivers into reaches separated by nodes.
2. Identifying where point sources (i.e. WwTW discharges) enter a river reach.
3. Delineating upstream catchments for diffuse sources (i.e. agriculture) and assigning these to river reaches.



4. Calculating the pollutant load from each source, which requires different feature engineering and data sources, depending on the source.
5. Calculating the in-stream concentration along the river network after inputs from each of the sources – also requires calculating flow for pollutant dilution.
6. Calculating a 'risk' value by comparing the concentration to established water quality standards.
7. Accounting for pollutant degradation from upstream to downstream, as we are dealing with non-conservative water quality variables that are affected by various in-stream water quality processes.

As with most models, there are limitations. Some of these are intrinsic to the model approach, whereas others which can be overcome by adaptations to the model or input of additional data.

- The model aims to provide an annual average estimation of risk across a catchment. Therefore, there is no representation of seasonal variation in risk, or risk due to rainfall events or low flow events, for example, besides how these affect the average risk.
- The approximation of river flow, needed to estimate in-stream pollutant dilution, is rudimentary. The validation exercise showed that estimations of risk by the model could be considerably improved by using measured rather than estimated flow. Integration of a complex flow estimation routine in the risk maps was not a priority, given that there is potential to bring in flow from other estimation methods, such as the [RDMAI Open Flow Model](#) or [QUBE](#) flows, for example.
- Estimation of loads from point sources, namely WWTWs and CSOs, is based on open data, primarily the consents database. This leads to a large degree of uncertainty, which can be reduced by incorporating water company data.
- Within the estimation of agricultural inputs of pollutants, livestock distribution is assumed to be uniform across a catchment.

The description and limitations of the RDMAI Risk Maps are given in far more detail in the model [documentation](#) on GitHub.

1.1.2 Overview of Existing Industry Model(s)/Tool(s)

SAGIS

SAGIS (Source Apportionment Geographical Information Systems) provides predictions of river water quality based on pollution source databases. The river quality model SIMCAT is used to predict annual flow and water quality statistics as annual means and percentiles in river catchments by including estimates of river flow and chemical input distributions to river reaches



for a set of 38 pollutants. Simple river chemistry processes of temperature dependent decay (applicable to phosphate) and interactions between BOD, Ammonia and Dissolved Oxygen are included. Results are presented as long-term predictions that can be interpreted as annual mean river flow and quality, with percentiles and monthly mean values.

The GIS element includes the spatial structure of a WFD river network, water bodies and input locations. The chemical source databases hold georeferenced loading data that are assigned to point (wastewater, industry) and diffuse sources (air deposition, agriculture, urban runoff, highways, mining etc.) and allocated as point or diffuse inputs for the rivers within a catchment area polygon. The loading contribution from the sectors and water quality predictions are reported along each river reach, catchment and at river basin district level.

SAGIS is widely used by the UK environmental regulators as a catchment planning tool for source apportionment to assess the relative inputs to rivers from wastewater, agriculture, industry and other sectors. This informs, at a river catchment scale, the reasons for failure of water quality standards and what actions might need to be taken.

The SIMCAT model was designed to be used to set permit levels for wastewater discharges with the statistical predictions set up to be directly compared to water quality standards. Confidence intervals based on the number of measurements in the input data are calculated to quantify uncertainty in the predictions. SAGIS is frequently called on to assess the impact of planned changes (developments, wastewater permit changes) on compliance with environmental quality standards.

As a tool used routinely by regulators, water companies and catchment partnerships, UKWIR coordinates SAGIS development, provides access to it and describes it as:

‘SAGIS is, however, most usefully conceptualised as a ‘system’ of interconnected tools and processes made up of mathematical models such as SIMCAT, MPER [Metal permitting] and RQP [single site permitting], the ArcGIS interface, data on discharge and water quality, river network and hydrology data, and external tools that rely on SAGIS input and output data, including calibration tools...’ [UKWIR](#), 2025.

The Environment Agency is the custodian of the English models and carries out updates to the model structure to update locations and maintains information about flow and quality inputs.



The water quality software tool SIMCAT is freely available under an open-source licence on [GitHub](#) where documentation includes the user manual¹. The GIS tool (SAGIS ArcPro system) and documentation is available to purchase from [UKWIR](#).

Limitations of SAGIS include data and the outputs. The databases require national level updates and as they are related to water company investment cycles, the data may not be representative of recent conditions. The load input databases are at a variety of scales from river reach to catchment area scale. This resolution may be too coarse for changes that the RDMAI risk map is aiming to capture. The outputs from SIMCAT are long-term annual statistics that may not predict short-term wet weather pollution.

The SAGIS outputs chosen for the benchmarking were the [Source apportionment of nutrient contributions to rivers in England and Wales modelled with SAGIS](#) (accessed November 2025, when the last update was April 2025). The dataset was produced using the Source Apportionment GIS (SAGIS) tool as a UK Water Industry Research Programme (UKWIR) project with support from the Environment Agency and SEPA using data from 2001 to 2012, depending on the data type.

This dataset provides estimates of loads and in-stream concentrations of nitrates and phosphates for England and Wales at a WFD waterbody catchment scale. Since the RDMAI Open Risk Map currently represents orthophosphate-P and ammoniacal-N, only the orthophosphate-P data from this SAGIS dataset are of relevance for the benchmarking (as nitrates are not directly comparable to ammonia). Pollutant inputs from various sectors are represented in this dataset. These include:

- Agriculture: arable and livestock farming. Agricultural phosphate losses were modelled with PSYCHIC based on 2010 census data.
- Highways: Modelled according to the Highways Agency Water Risk Assessment Tool (HAWRAT) based on average annual rainfall, highway network maps, and mean event concentrations reported by the Highways Agency.
- Urban runoff: Calculated according to rainfall intensity, CORINE land use data and mean event concentrations reported in the literature.
- Wastewater treatment works (WwTWs): calculated using the EA water quality samples database (WIMS) and flow data for WwTWs reported to the EA by water companies (MCERTS - both 2010-2012 inclusive).

¹ Tony Warn, The SIMCAT Manual, 2018 [simcat/docs/181118_SIMCAT_Manual.docx_at_main · DEFRA/simcat · GitHub](#)



- Combined sewer overflows (CSOs): Calculated considering CSO and storm tank locations, chemical monitoring data from the Chemical Investigations Programme (CIP), rainfall intensity, and CORINE land use data.

For various points in each water body catchment, the dataset includes mean phosphate concentration (mg/l), and associated upper and lower confidence limits, as well as a phosphate concentration value for each of the pollutant inputs represented.

1.2 Catchment Selection

The selection of catchments for benchmarking aimed to be representative of the following characteristics across England:

- The different regions of England.
- Differences in annual rainfall, as this affects non-point sources as well as intermittent discharges.
- Differences in urban area extent, as an indication of relative contributions of point and non-point sources.

Table 1.1 shows a summary of the main characteristics of the chosen benchmark catchments, while Figure 1.1 shows their positions within the United Kingdom. It should be noted that the upper reaches of the River Wye are in Wales, whereas the lower reaches are in England. This benchmarking includes only those sub-catchments of the River Wye that are entirely in England.

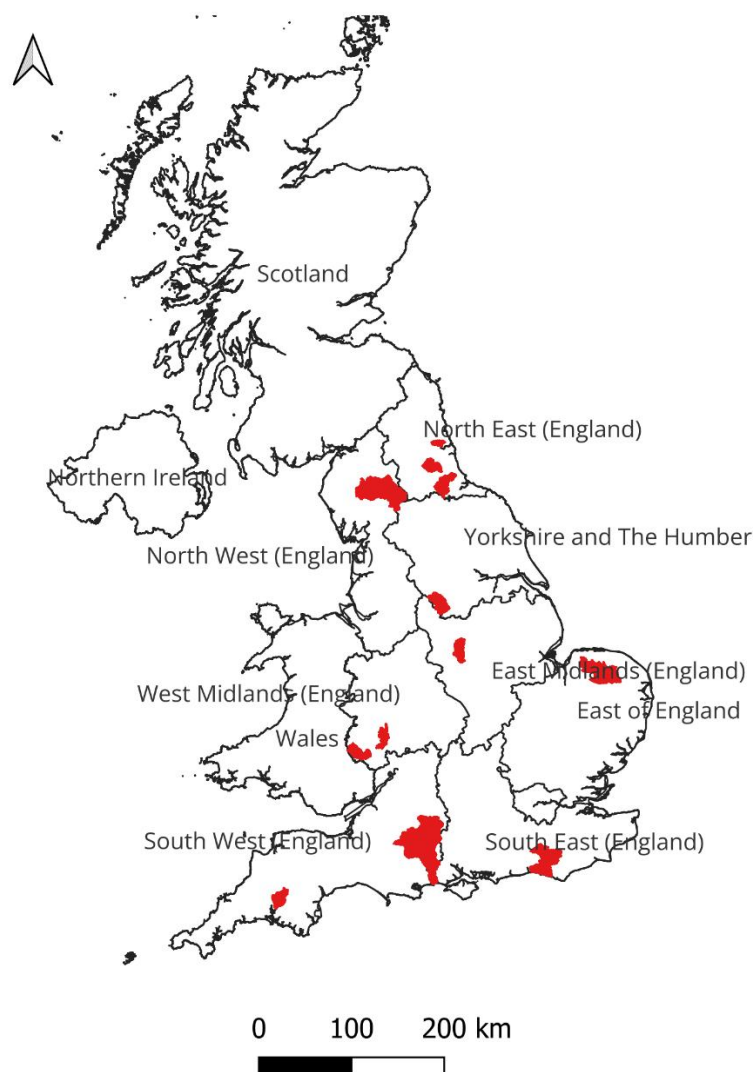
Table 1.1 Summary of catchments chosen for benchmarking the RDMAI Open Risk Map

Region	Catchment	Annual Rainfall (SAAR; mm)	Catchment area (km ²)	Urban extent (%)
North East	Ouse Burn	668	61.6	58%
North East	Browney	932	182.1	7.5%
North East	Skerne	654	259.6	20.1%
North West	Eden Upper & Eamont	1142	1083.1	1.63%



Region	Catchment	Annual Rainfall (SAAR; mm)	Catchment area (km ²)	Urban extent (%)
Yorkshire and Humber	Upper Don	1131	269.7	9.0%
East Midlands	Erewash River	708	207.6	33%
West Midlands	Sections of River Wye in England	706	392.5	0.84%
East England	Wensum	672	659.8	5.9%
South East England	Upper & lower Ouse	835	635.6	14.8%
South West	Hampshire Avon	782	1750	4.8%
South West	Tavy	1544	206.5	3.36%

Figure 1.1 Catchments chosen for benchmarking the RDMAI Risk Maps



1.3 Benchmarking Methodology

Please note, while the RDMAI Open Risk Map provides estimates of risk for orthophosphate-P and ammoniacal-N, benchmarking was only done for orthophosphate-P, due to ammoniacal-N not being represented in the UKWIR SAGIS dataset. Therefore, when describing the benchmarking methodology below, only data from the RDMAI Open Risk Map relating to orthophosphate-P are mentioned.

The RDMAI Open Risk Map tool was run for each of the benchmark catchments (Table 1.1, Figure 1.1). The script for running the model, as found in the [RDMAI Github](#) repository (accessed November 2025), was adapted to output the nodes of the network as a shape file. Within this shapefile were attributes for each node, including total orthophosphate-P risk and the risks attributable to each 'pressure' or pollutant source represented in the risk map, namely

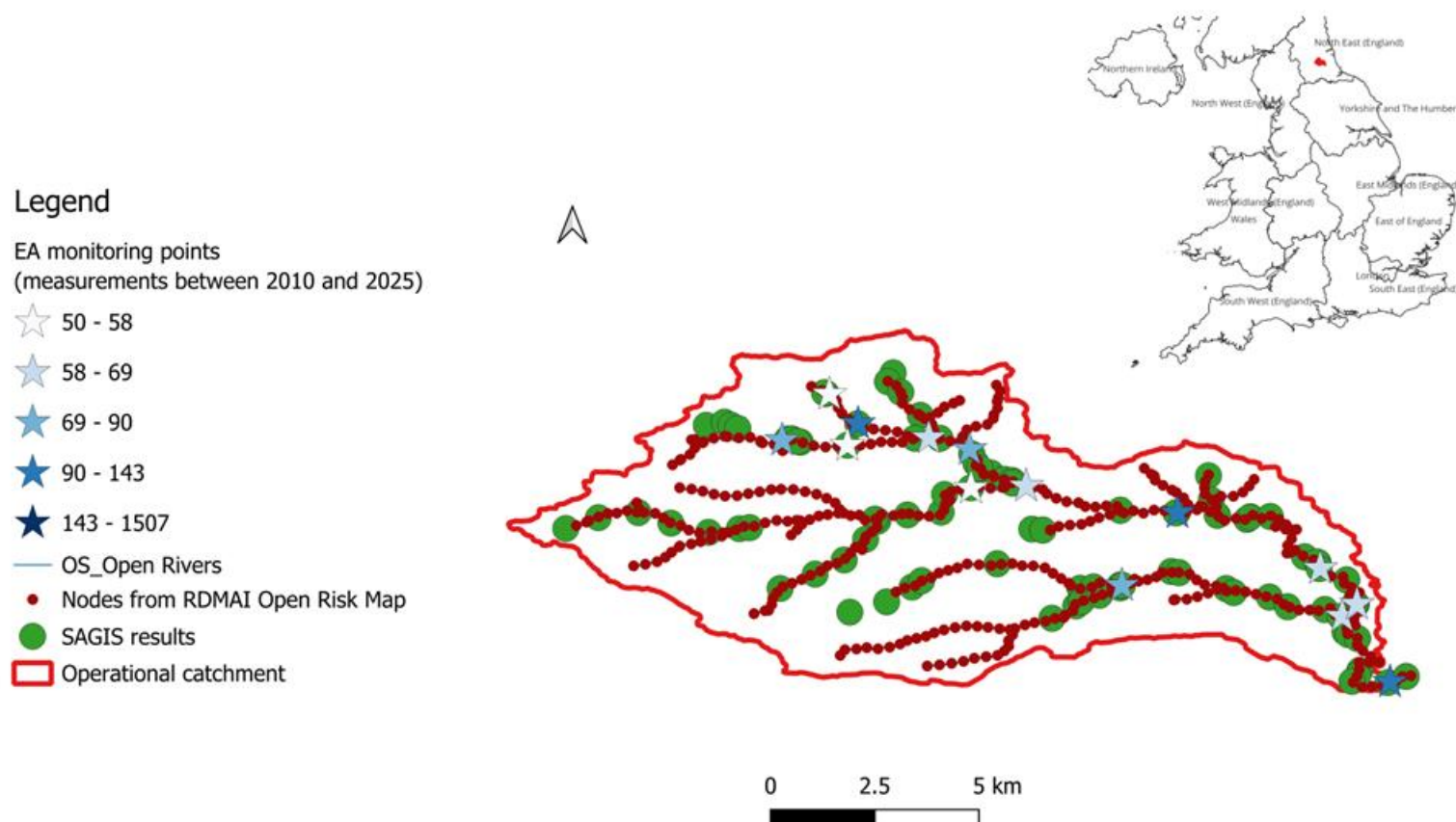


WwTWs, intermittent discharges (CSOs), agriculture, road runoff, and urban runoff. Then, once the shapefiles were extracted, they were modified to calculate the orthophosphate-P concentrations associated with each of the risks based on the orthophosphate-P standard used (average of the UKTAG orthophosphate-P standards for a river to be classified as 'Poor' across the different altitude and alkalinity categories).

These risk map shapefiles were then plotted in GIS alongside the UKWIR SAGIS output polygons. The Environment Agency water quality monitoring database (WIMS) for 2010 to 2025 was examined, and any monitoring points for running river water containing at least 50 measures of “Orthophosphate, reactive as P” within the catchment were identified and also plotted. For each EA water quality monitoring point, the mean, standard deviation and number of orthophosphate-P samples (n) were calculated. As an example, Figure 1.2 shows the datasets plotted together for the Browney catchment in North-East England.



Figure 1.2 Map of the Browney catchment in North-East England showing nodes from the RDMAI Risk Map, nodes from the UKWIR SAGIS dataset, and Environment Agency water quality monitoring points (for running river water) containing at least 50 measures of orthophosphate-P (from 2010 to 2025)





Associating an UKWIR SAGIS node with an RDMAI Risk Map node

The task was to associate an UKWIR SAGIS node with an RDMAI Risk Map node, so that the orthophosphate-P estimates by each model could be compared. Since there were a lot of nodes for the RDMAI Risk Map and UKWIR SAGIS dataset in each catchment, this association was not done manually. Instead, a script was created to identify this association: if there was no RDMAI Risk Map node within 50 m of a UKWIR SAGIS node, an association was not made. Each SAGIS node was matched with one Risk Map node.

Using these node associations, the mean orthophosphate-P concentrations and contributions from each pollutant source, calculated from each model, could be compared. In this way, the overall orthophosphate-P concentration calculated by the UKWIR SAGIS model (along with the upper and lower confidence intervals) could be compared to that calculated by the RDMAI Risk Map, and a comparison of the concentration of orthophosphate-P from each pollutant source calculated by the two models could identify differences between each of the models' outputs. Please note, the ordering of paired points in subsequent graphs is according to ascending UKWIR SAGIS estimated orthophosphate-P, and is not related to catchment placement.

Associating UKWIR SAGIS-RDMAI Risk Map node pairs with an EA water quality monitoring point

While a comparison of the orthophosphate-P estimates between the two models can highlight differences, it does not give an indication of model accuracy. Therefore, EA water quality monitoring points within each catchment, measuring running river water, and containing at least 50 measures of orthophosphate-P between 2010 and 2025, were identified and plotted in GIS. Then, associations between these EA monitoring points and UKWIR SAGIS-RDMAI Risk Map node pairs were done manually by zooming into a river reach. Only if the model node pairs and the EA water quality monitoring point were within 300 m were the associations made. The choice to make this association manually was made due to:

- The relatively limited number of qualifying EA water quality monitoring points.
- To avoid making an association if the model node pair and the EA water quality monitoring point were on different branches of the water body – this is difficult to automate.

Making this comparison allowed benchmarking of orthophosphate-P estimates by both the UKWIR SAGIS and RDMAI Risk Map models with monitored orthophosphate-P data. Please note, in the graphs showing the comparisons, the ordering of points is according to ascending EA mean measured orthophosphate-P, and has nothing to do with the placement of nodes in the catchment. For example, points closest to the Y- axis should not be interpreted as representing nodes in the upper catchment.



1.4 Benchmarking Limitations

A major limitation in the benchmarking is that it can only be conducted for orthophosphate-P when benchmarking against the UKWIR SAGIS dataset. This is because the UKWIR SAGIS dataset does not include estimates of ammoniacal-N.

However, orthophosphate-P is suitable for benchmarking as the important pollution sources represented in both the UKWIR SAGIS and RDMAI Risk Map all contribute to instream orthophosphate-P, including agricultural diffuse pollution.

Another possible limitation is the identification of UKWIR SAGIS-RDMAI Risk Map node pairs. This was automated due to the large number of nodes in each catchment. It is possible that some pairings may be invalid as the nodes may fall on different branches of the river network – in this case, we can expect loads of pollutants coming from upstream to be significantly different between the paired nodes. An attempt to minimise this possibility has been made by considering a 50-m distance threshold in the node pairings.

When comparing UKWIR SAGIS and RDMAI Risk Map estimates of orthophosphate-P to EA monitoring point average measures, there were limited associations that could be made in each benchmark catchment. To achieve a reasonable long-term indication of observed orthophosphate-P, only monitoring points with 50 or more measures between 2010 and 2025 were considered – this also allowed the calculation of reasonable confidence limits in the observed mean. This association between UKWIR SAGIS points, RDMAI Risk Map points and EA monitoring points also did not assess for the presence of pollution sources between the points, such as WwTW or CSO outfalls. It was assumed that the threshold of 300 m distance between the points would limit pollution sources falling between the points.

It is noted that potential changes in pollution inputs are likely to have occurred between 2010 and 2025 – the period chosen to investigate EA water quality monitoring data. These include how often CSOs operate due to population growth and more intense rainfall (and subsequently any remediation undertaken by water companies), and changes to treated effluent permits (which are increasingly stipulating phosphorus treatment). Therefore, the calculated mean orthophosphate-P for EA water quality monitoring points over the 15-year period may not reflect more recent changes to orthophosphate-P loading.

The benchmarking exercise can note differences in estimated orthophosphate-P between the UKWIR SAGIS model and the RDMAI Risk Map. However, these differences can be due to differences in data inputs and/or feature engineering of pollution inputs. Since it was not possible to align the data inputs of the two models, identifying the main source of model differences (data inputs or feature engineering) is not possible.

The RDMAI Risk Map incorporates the latest available open datasets, whereas updates to the UKWIR SAGIS datasets are done periodically; therefore, an updated UKWIR SAGIS database may show a different benchmarking performance than observed in this report.



SAGIS represents highway runoff using the Highways Agency Water Risk Assessment Tool (HAWRAT). Therefore, SAGIS only represents road runoff pollution from highways and A-roads. Urban and rural road runoff contributions represented in the RDMAI Open Risk Map are from trunk roads (roads within 200 m of a river reach). Therefore, when comparing the estimated highway/road runoff orthophosphate-P contributions from the two models, it must be considered that they are representing these contributions very differently.

While this benchmarking exercise does attempt to include catchments that are reasonably representative of the different regions of England, the benchmark catchments are still relatively limited. A more comprehensive benchmarking including a larger number of catchments may yield different conclusions.

A separate report (available on [GitHub](#)) outlines the validation of the RDMAI Risk Map against orthophosphate-P measured at EA water quality monitoring points and against frequent sonde measurements of ammonium-N. In the validation, the RDMAI Open Risk map showed a general over-estimation compared to orthophosphate-P measured at EA water quality monitoring points, and a general under-estimation compared to frequent sonde measurements of ammonium-N. This difference could be due to differences in the way the RDMAI Risk Map represents the two pollutants. However, it is more likely that the differences are due to the frequency at which the two pollutants were measured. EA water quality measures occur infrequently, on average between once or twice a month. Sonde measures are frequent, between 15 minutes to an hour. Therefore, the EA measurements may miss spikes in orthophosphate-P (particularly as during storm events, health and safety considerations may prevent sampling). This would push the calculated mean of EA measures down. In contrast, the more frequent sonde measures would capture spikes in ammonium-N, acting to push the average measure up. This limitation should be considered when comparing RDMAI Risk Map estimates of pollutant concentrations to the averages of EA water quality monitoring data.



2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2.1 Comparison of model 'ease-of-use'

Model Outputs	SAGIS	RDMAI Model
Installation & Run Process	The SAGIS system requires expert water quality and ArcGIS users to install and run the models. The set up is not easy as it is an extensive set of files to connect to the source apportionment databases. There are associated spreadsheets to edit inputs, calibrate the model and view the model results that take time to learn to use. The user guides are comprehensive as there are a lot of detailed settings to review. Email support is available from the EA and consultants who update the SAGIS system.	Running the RDMAI Risk Maps requires basic Python knowledge: How to clone a GitHub repository, how to set up a Conda environment, and how to use Jupyter Notebooks. The RDMAI Risk Maps are 'plug-and-play'. Once the repository is established, running an existing notebook will generate the risk map for the catchment of interest.
Data inputs	The source apportionment data and model structures are already set up for the user. There are instructions on editing the model structure and inputs as required. This involves setting up summary statistics (e.g. mean phosphate for wastewater treatment works) or seasonal distributions (e.g. monthly diffuse phosphate loads from agriculture).	The EA consents database needs to be manually downloaded into a stipulated folder. All other datasets are processed automatically by the provided script. Rivers: Open River Network Overflow spill data: EA Storm Overflows - Annual Returns



	The data updates are limited to periodic updates coordinated around the 5-year water company investment cycle and the EA requirements to use SAGIS results for the 6-year River Basin Planning and reporting cycle.	Consents data: EA Consents Database Land use: CROME 2023 dataset Hydrology of Soil Types (HOST) derived from this study. Rainfall data: UK Met Office Rainfall Data Roads and buildings: OS OpenMap - Local Built-up areas: OS Open Built Up Areas Farmscoper Data: ADAS Farmscoper Tool is used for agricultural and livestock runoff estimates via lookup tables.
Licensing Requirements	A license is required to use SAGIS system. The SIMCAT software can be used with an open-source licence independently of SAGIS for individual catchments. The SAGIS report and tool are available to purchase from UKWIR for £600 (+ fees for additional data licencing). It is available through UKWIR to water companies. The Environment Agency have made some outputs available through the CaBA Data Package.	All data used in the RDMAI Risk Map is open and free to use.



2.2 Model Outputs

Table 2.2 Comparison of model outputs

Model Outputs	SAGIS	RDMAI Model
Data Format	Results from SAGIS are both in SIMCAT format (sets of text files and CSV files with georeferencing of water quality predictions and source apportionment at each model result point) and these results files are imported into the SAGIS ArcGIS Pro viewer and viewed as shapefiles. Results relevant to this comparison are long-term phosphate predictions as mean values with confidence intervals.	The RDMAI Risk Map produces an interactive map in a Jupyter Notebook showing the catchment as a series of reaches separated by nodes. Clicking on a node shows you the total risk for orthophosphate-P and ammoniacal-N at that node and the relative contributions in risk from the different pollution sources at that point. The Jupyter Notebook can be easily adapted to output the risk map as a GIS shape file.
Temporal Resolution	SAGIS provides annual average concentrations for 38 pollutants including phosphate. Results include source apportionment, confidence limits around the mean and percentiles. Comparisons to water quality standards are also provided.	The RDMAI Risk Map provides annual average estimates of orthophosphate-P and ammonia concentration, represented as risks.



Geographic Constraints	Model results are available for all non-tidal WFD river reaches for the UK.	Some of the datasets the RDMAI Risk Map depends on, such as the EA consents database, are constrained to England. Therefore, the Risk Maps can only be applied to England at this stage.
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2.3 Time and Cost Comparison

Although the SIMCAT software is free to use, the SAGIS model version can only be used by experienced water quality modellers and requires access to ArcGIS Pro and purchase of the SAGIS tools. The EA do provide on request truncated versions of the SIMCAT model files (.DAT) for specific river catchments that can be used with the free software. These files would include the source apportionment inputs set up by the EA to represent diffuse and point loads.

The RDMAI Risk Map is open and free to use, as are the input data. The model is virtually 'plug-and-play', so is relatively easy to set up and run. Running a catchment, depending on the size of a catchment, can take between 5 and 30 minutes on a standard laptop.

2.4 Model Performance

The results presented below are separated by benchmarked catchment. For each benchmark catchment and for each model node pairing, the RDMAI Risk Map estimate of annual average orthophosphate-P concentration is compared to that by the UKWIR SAGIS model. Since the UKWIR SAGIS model provides confidence intervals for mean orthophosphate-P, it is represented as a point with whiskers. In contrast, the RDMAI Risk Map estimate is represented as a point. Therefore, if the RDMAI Risk Map estimate falls within the whiskers, then it is concluded that the models give the same overall result. Importantly, the points in the graphs are in order of increasing UKWIR SAGIS concentration, so the ordering is not related to the positions of the node pairings in the catchment.

Also represented for each benchmark catchment is a model comparison of orthophosphate-P concentration attributable to each pollutant source. This is represented as an X-Y scatter plot, with each point representing a model node pair. The scatter plots provide a broad indication of differences in scale between how the two models represent a particular pollution source.

Finally, the mean orthophosphate-P values estimated by the model node pairs are compared to the mean observed EA monitoring point orthophosphate-P. The confidence interval for the EA monitoring point mean orthophosphate-P was calculated as:

$$\mu \pm Z \times \left(\frac{\sigma}{\sqrt{n}}\right)$$

where μ is the sample mean, σ is the sample standard deviation, n is the number of samples, and Z is the Z-value corresponding to a 95% confidence (1.96).

The EA water quality measures of orthophosphate-P are then represented as points (means) with whiskers (showing the confidence limits), whereas the UKWIR SAGIS and RDMAI Risk Map estimates of orthophosphate-Ps are shown as points. Model estimates falling within the whiskers then represent accurate model estimates. Importantly, the points in the graphs are ordered by increasing EA measured orthophosphate concentration, with the ordering not related to positions of the node pairings in the catchment. In addition, an X-Y scatter plot of

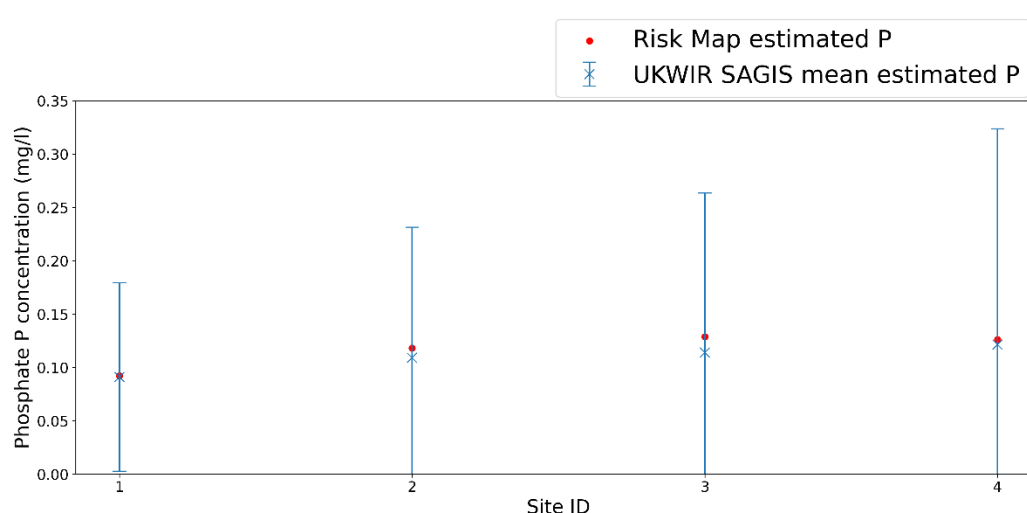


model estimated versus mean EA measured orthophosphate-P is shown for each benchmark catchment. For each benchmarked catchment, catchment characteristics, including Standard Annual Average Rainfall (SAAR), are shown in Table 1.1.

2.4.1 Ouse Burn

The Ouse Burn is a small but highly urbanised catchment, with a relatively low rainfall.

Figure 2.1 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Ouse Burn. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



As shown in Figure 2.1, RDMAI Risk Map estimates of orthophosphate-P concentration were close to that of the UKWIR SAGIS model for the Ouse Burn and concentrations were relatively low.

Figure 2.2 shows that, for the Ouse Burn:

- The RDMAI Risk Map model estimates small concentrations of WwTWs, possible related to one or more private WwTW consents, whereas the UKWIR SAGIS model estimates none.
- Both models estimate relatively small concentrations from CSOs, although the Risk Map estimate is higher.
- Both models estimate small concentrations from crop agriculture.
- Both models estimate small concentrations from livestock.



- Estimations of orthophosphate-P from urban runoff and highways were low for both models, although the Risk Map estimates are higher for highways and the UKWIR estimates of urban runoff are higher.

Figure 2.2 RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Ouse Burn

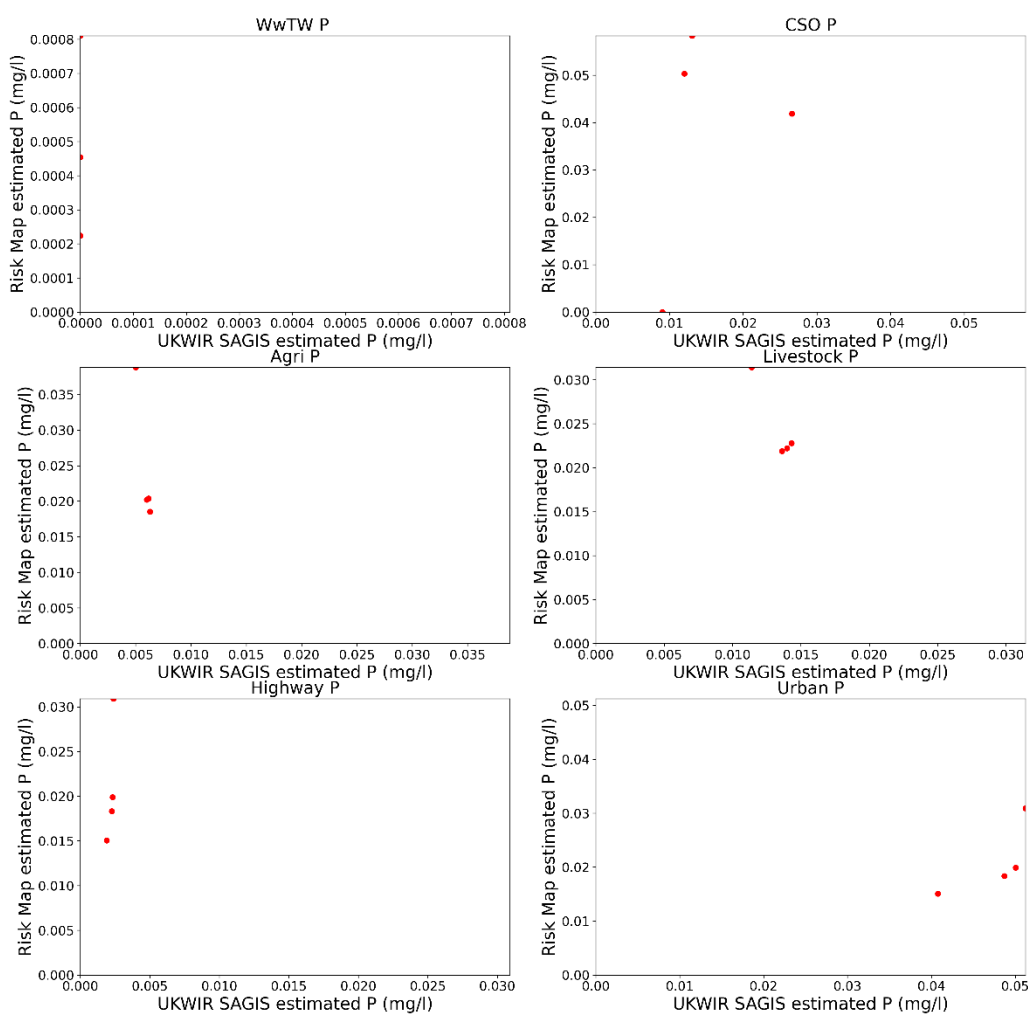


Figure 2.3 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Ouse Burn - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

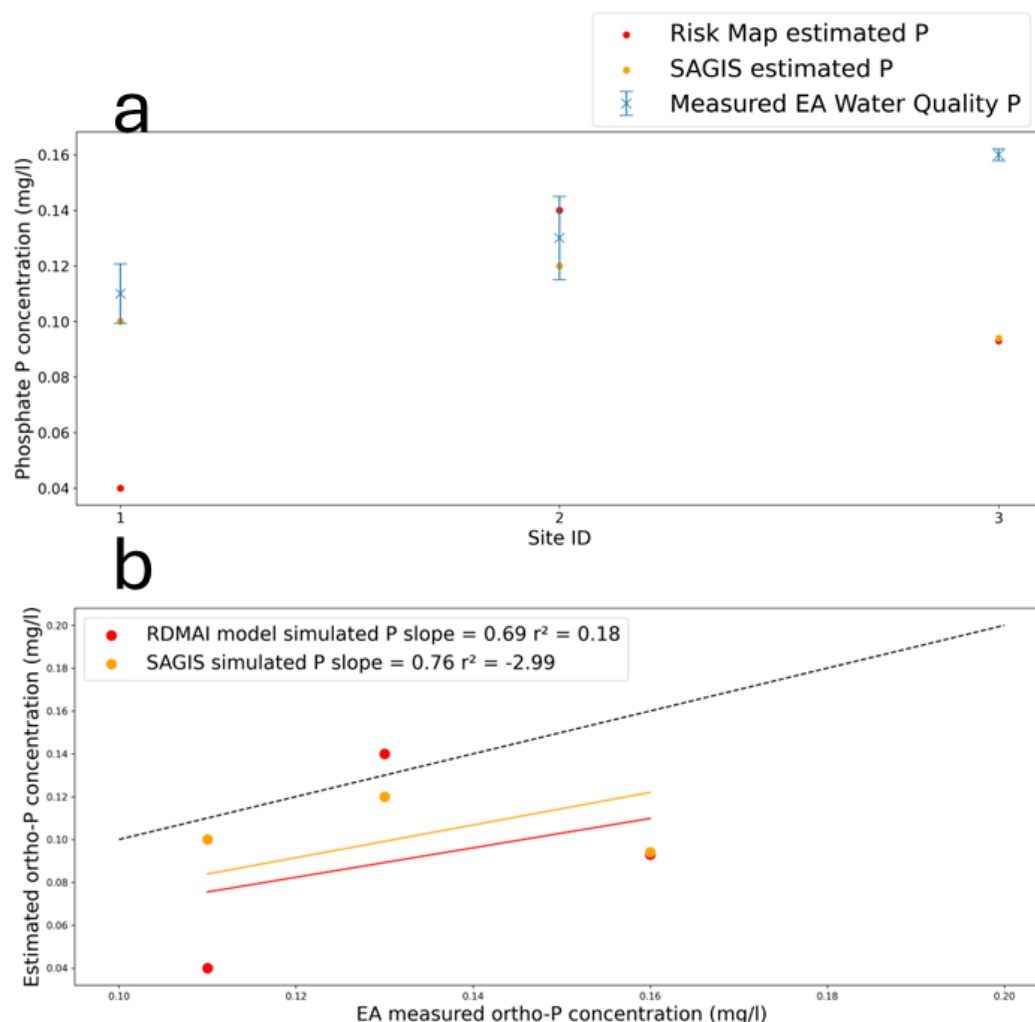


Figure 2.3(a) shows that both the UKWIR SAGIS and RDMAI Risk Map estimates of orthophosphate-P were low compared to EA measured orthophosphate-P for sites 1 and 3. For site 2 the SAGIS estimate fell within the confidence limits. Figure 2.3(b) shows that both models have a weak relationship to the measured orthophosphate-P. However, given the limited number of comparison sites (3), conclusions are tentative.

2.4.2 Browney

The Browney catchment has a relatively higher rainfall among the benchmark catchments (see Table 1.1), but is not very urbanised.



Figure 2.4 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Browney. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P

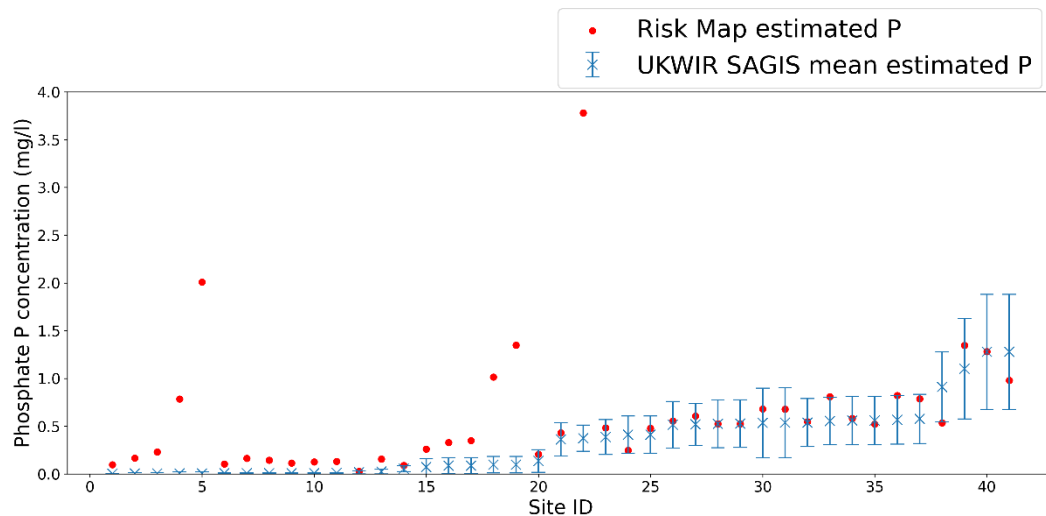


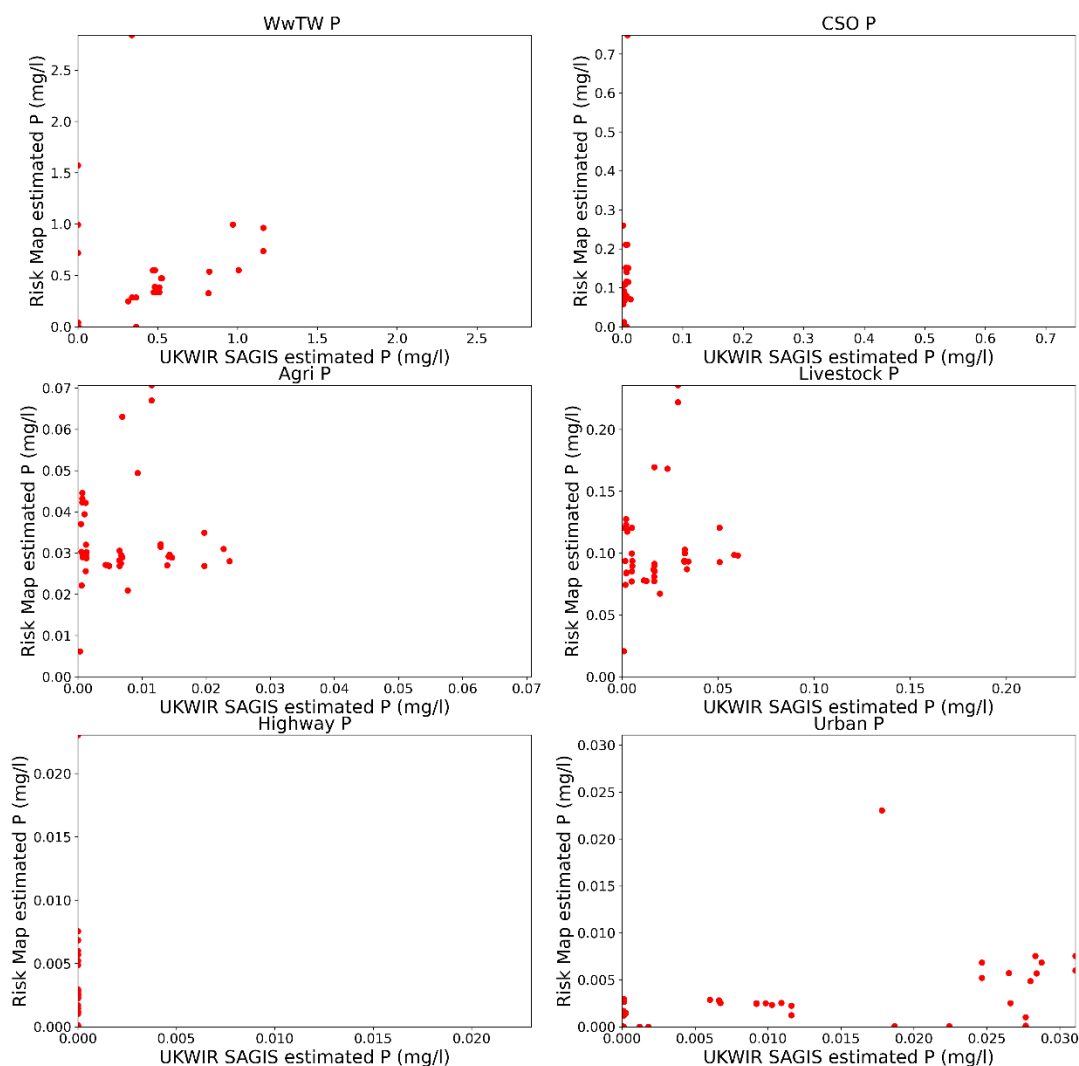
Figure 2.4 shows that the RDMAI Risk Map gives similar orthophosphate-P concentrations to the UKWIR SAGIS model for the Browney for sites with overall higher UKWIR SAGIS model concentrations of orthophosphate-P. For sites with lower UKWIR SAGIS model estimates, the RDMAI Risk Map estimates are higher. For only a few sites do the Risk Map and SAGIS disagree strongly (e.g. sites 5, 19, 22).

As shown in Figure 2.5, for the Browney:

- Estimates of WWTW concentrations of orthophosphate-P are relatively consistent between the two models.
- The RDMAI Risk Map provides estimates of CSO orthophosphate-P concentrations while those of the UKWIR SAGIS model are absent or negligible.
- The RDMAI Risk Map estimates of agricultural orthophosphate-P exceed those by the UKWIR SAGIS model.



Figure 2.5X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Browney

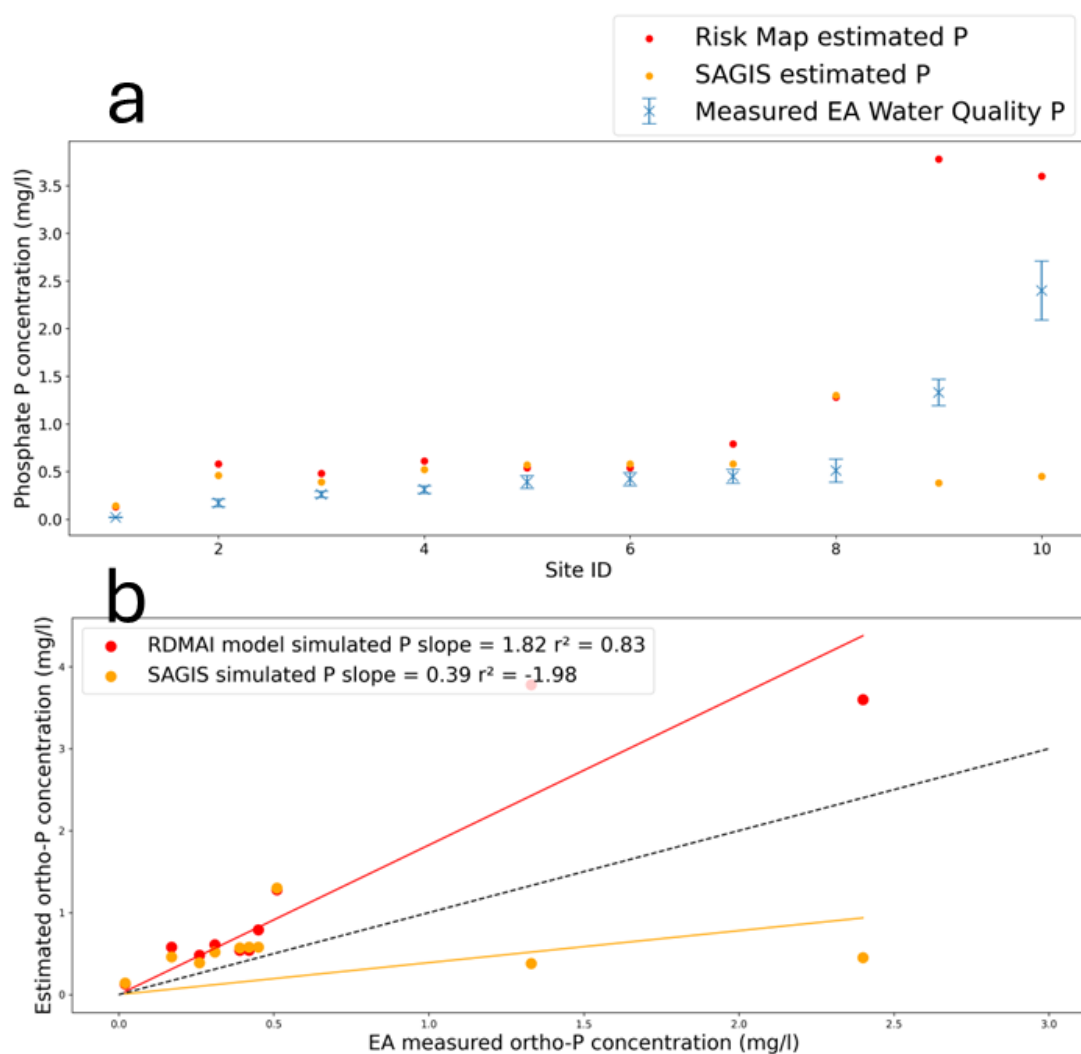


- The RDMAI Risk Map estimates some highway runoff and the UKWIR SAGIS model does not, likely due to differences between the two model in the way they represent road concentrations of pollutants.
- The UKWIR SAGIS model estimates of urban runoff P exceed that of the RDMAI Risk Map.

As shown in Figure 2.6(a), both the RDMAI Risk Map and UKWIR SAGIS model estimates of orthophosphate-P are similar to that of observed orthophosphate-P for all sites except those with the extreme highest measured orthophosphate-P concentration, where the Risk Map estimates are too high and the UKWIR SAGIS estimates are too low (sites 9 and 10). Figure 2.6(b) shows that the RDMAI Risk Map has a stronger linear relationship with the measured orthophosphate-P than the UKWIR SAGIS model for this catchment.



Figure 2.6 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Browney - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

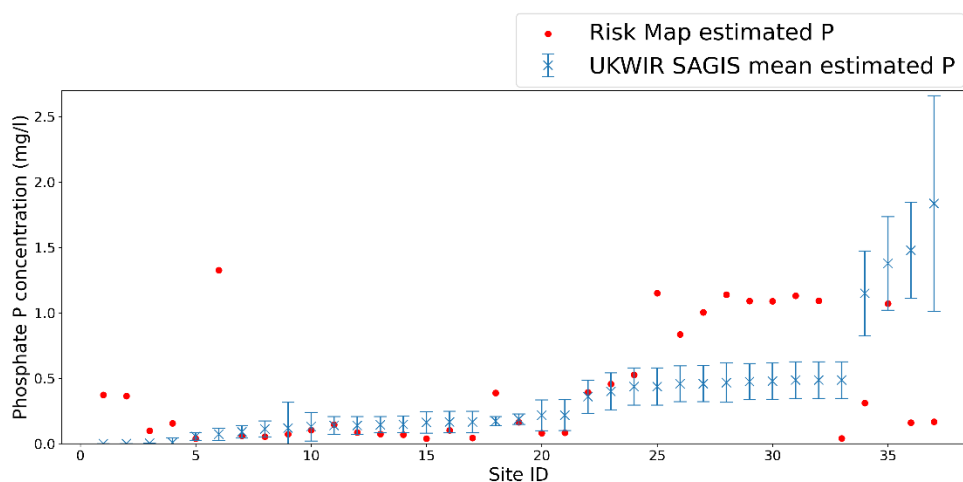


2.4.3 Skerne

The Skerne catchment is urbanised and has a relatively low rainfall.



Figure 2.7 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Skerne. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



As shown in Figure 2.7, the RDMAI Risk Map estimates of orthophosphate-P tended to be similar to that of UKWIR SAGIS for low values (i.e. < 0.5 mg/l) but exceeded the UKWIR SAGIS estimates for intermediate values (e.g. sites 25 to 32) and then were lower than the SAGIS estimates at the extreme higher values (e.g. sites 33, 34, 36, 37).

When examining the individual contributors to differences in Figure 2.8, it is clear that:

- UKWIR SAGIS estimates of WWTW concentrations of orthophosphate-P were similar to those of the RDMAI Risk map.
- The RDMAI Risk map tended to generate higher estimates of CSO concentrations.
- Besides for a single site, estimates of orthophosphate-P were similar for the two models.
- The RDMAI Risk Map simulates a greater contribution of orthophosphate-P for highways, whereas both models generate similar urban runoff concentrations for some sites. SAGIS provided estimates of 0 mg/l orthophosphate-P for both highways and urban areas for some sites, whereas the Risk Map generally returns (an albeit low) value. It should be noted at this point that urban and highway run-off are not thought to be significant contributors to orthophosphate-P pollution and mechanisms for this pollution pathway may not be fully understood and hence modelled.



Figure 2.8 RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Skerne

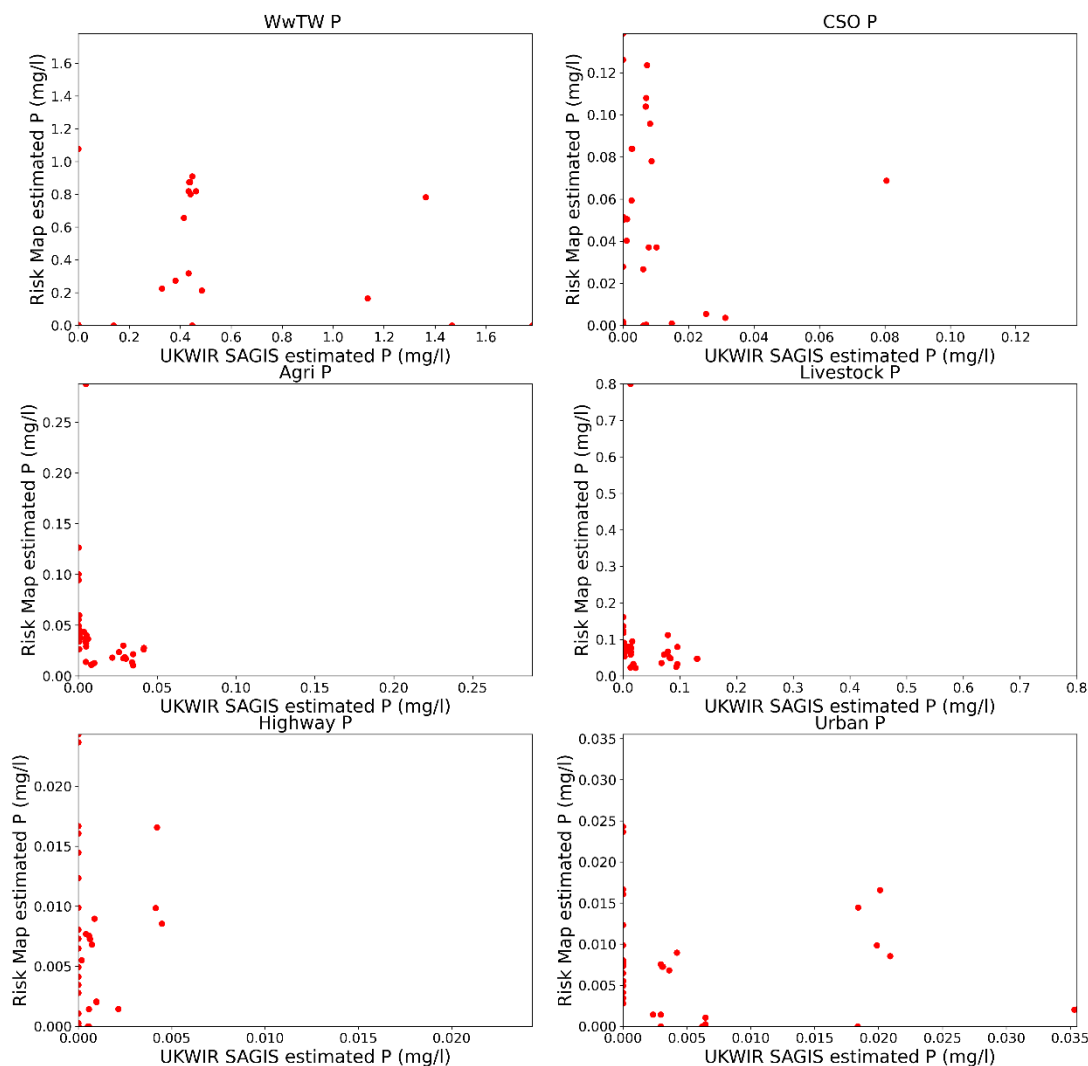
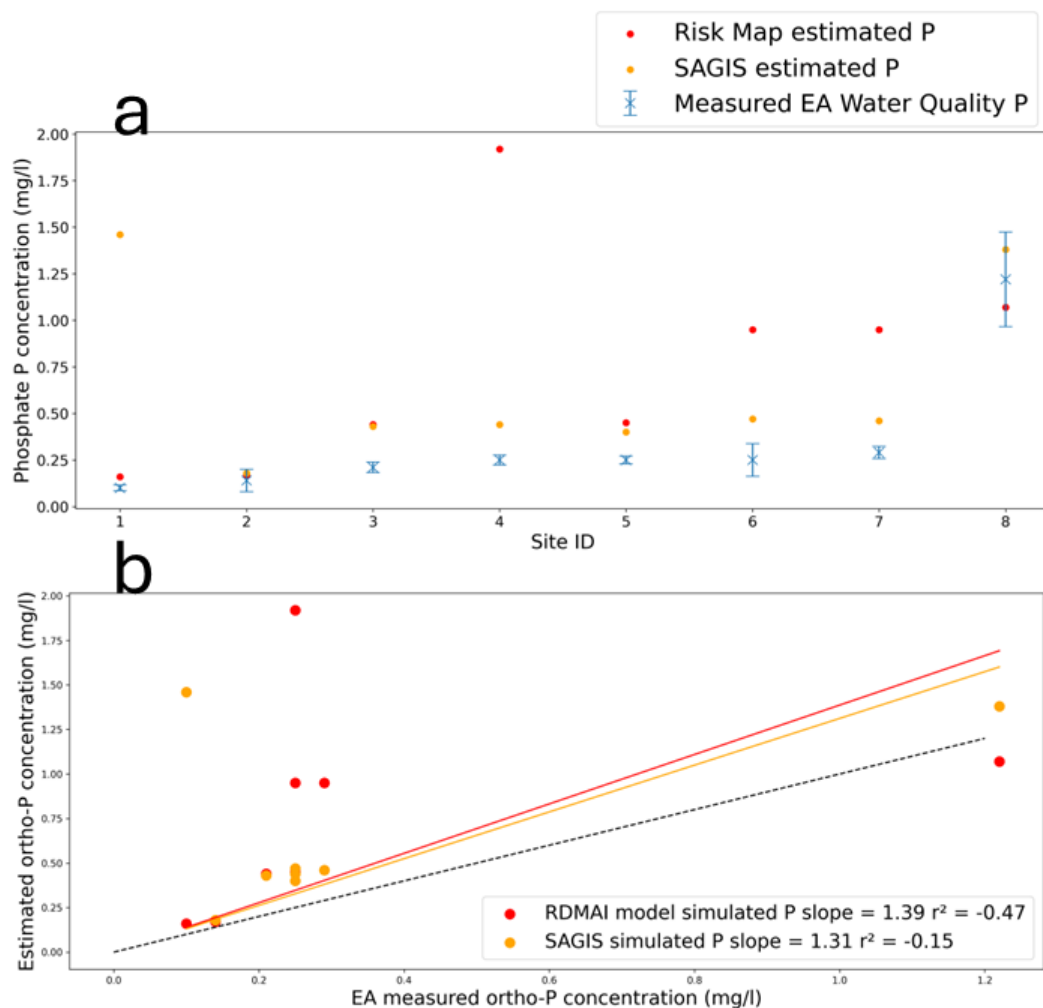


Figure 2.9 (a) shows that both the UKWIR SAGIS and RDMAI Risk Map tend to over-estimate orthophosphate-P relative to the EA water quality monitoring, although the overestimation by the RDMAI Risk Map is higher, except for site 1. Figure 2.9 (b) shows that both models tend to have a poor linear relationship with measured orthophosphate-P, and overestimate overall. Looking at Figure 2.8, it is likely that the representation of WwTW orthophosphate-P contributions by the two models is responsible for the major differences to mean EA measured orthophosphate-P.



Figure 2.9 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Skerne - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.



2.4.4 Upper Eden

The Upper Eden and Eamont catchments are rural and large catchments with a relatively high rainfall.



Figure 2.10 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Upper Eden. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P

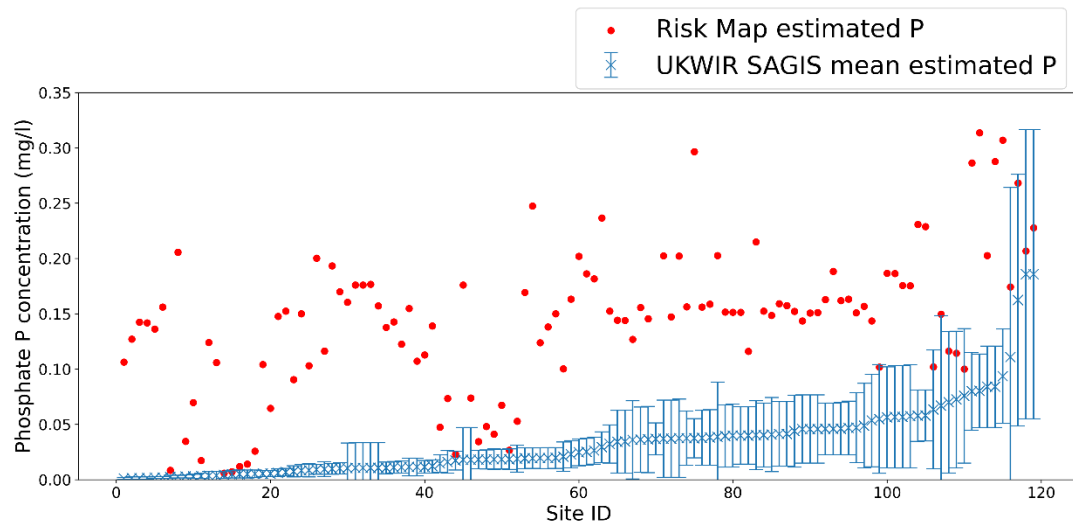
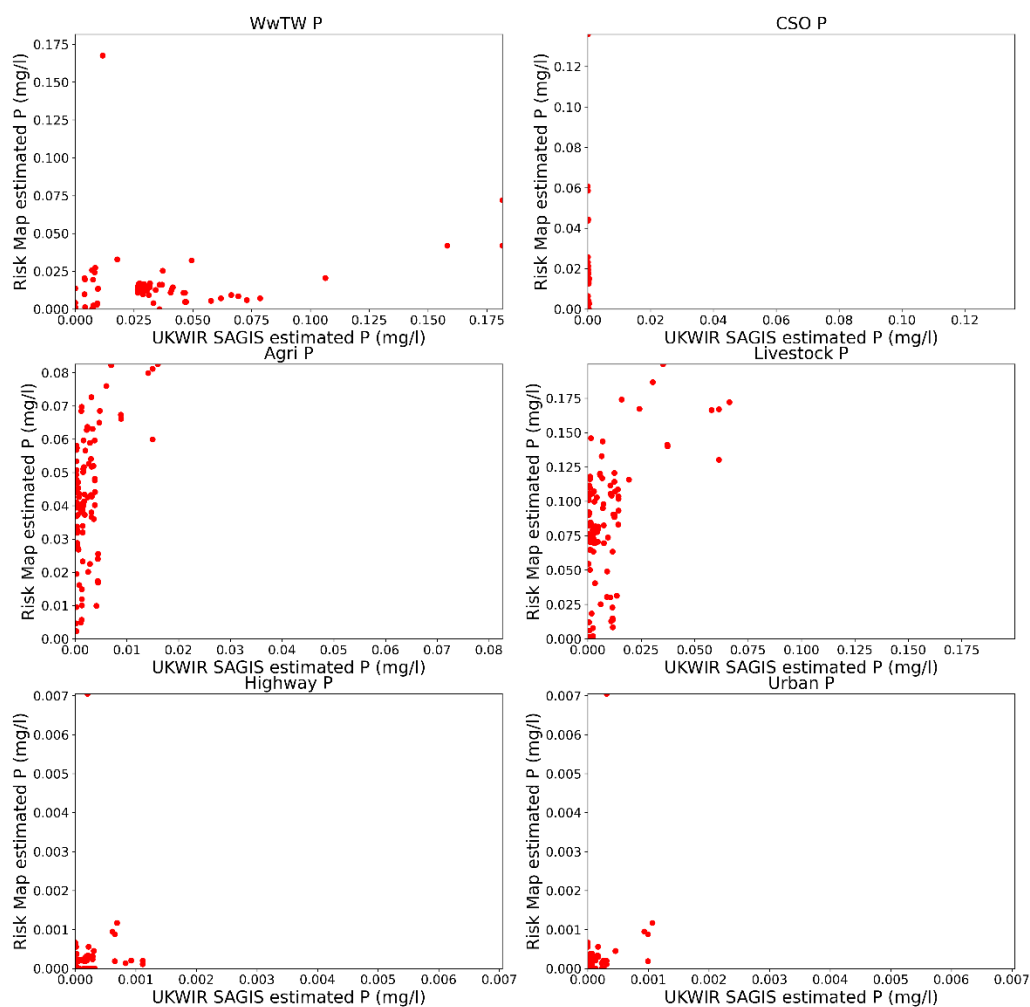


Figure 2.10 shows that the RDMAI Risk Map consistently over-estimates orthophosphate-P relative to the UKWIR SAGIS model for the Upper Eden. When examining the cause of differences in Figure 2.11, we can see that:

- WwTW estimates of orthophosphate-P by the UKWIR SAGIS model exceed that by the RDMAI Risk Map.
- The RDMAI Risk Map provides CSO concentrations of orthophosphate-P while the UKWIR SAGIS model provides zero or negligible concentrations.
- The RDMAI Risk Map estimates of orthophosphate-P for crop agriculture and livestock exceed that by the UKWIR SAGIS model.
- Estimates of highway and urban concentrations of orthophosphate-P between the two models are relatively consistent, besides for a single site.



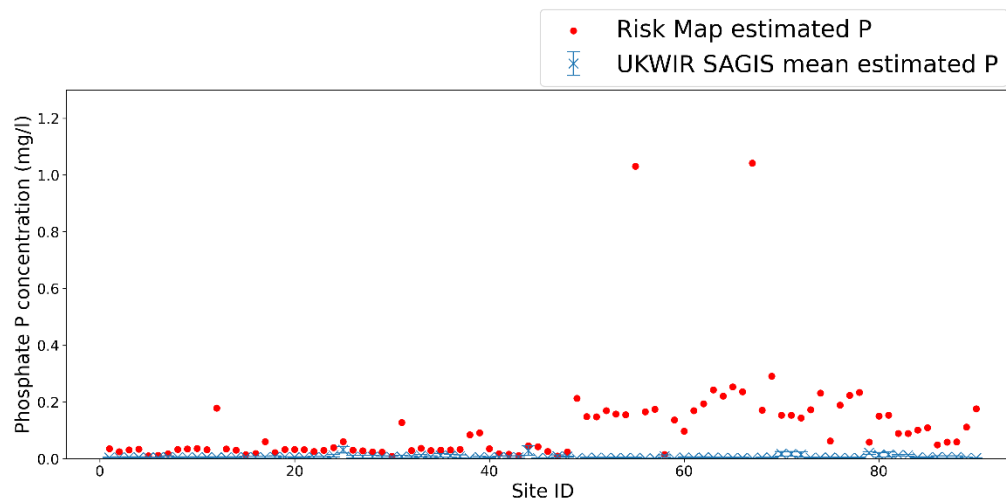
Figure 2.11 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Upper Eden



2.4.5 Eamont



Figure 2.12 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Eamont. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



As shown in Figure 2.12, estimates of orthophosphate-P by the UKWIR SAGIS model for Eamont were relatively low, and the RDMAI Risk Map estimate tended to be higher for most sites. When examining the reasons for differences in Figure 2.13, it is clear that:

- The compared models showed consistent concentrations of WwTW orthophosphate-P.
- The RDMAI Risk Map provided some minor CSO contribution of orthophosphate-P, whereas concentrations from the UKWIR SAGIS model were negligible or absent.
- The RDMAI Risk Map provided estimates of crop agriculture and livestock concentrations of orthophosphate-P, whereas those by the UKWIR SAGIS model were negligible or absent. It is this difference that is mostly responsible for the higher RDMAI Risk Map estimates shown in Figure 2.12.
- Both models showed negligible contributions of highways and urban runoff to orthophosphate-P concentration that were relatively consistent.

As shown in Figure 2.14(a), the RDMAI Risk Map estimates of orthophosphate-P were far higher than the EA water quality monitoring observations, whereas the UKWIR SAGIS estimations were far closer to the EA observations. However, as shown in Figure 2.14(b), the RDMAI Risk Map showed a better trend relationship to measured orthophosphate-P.



Figure 2.13 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Eamont

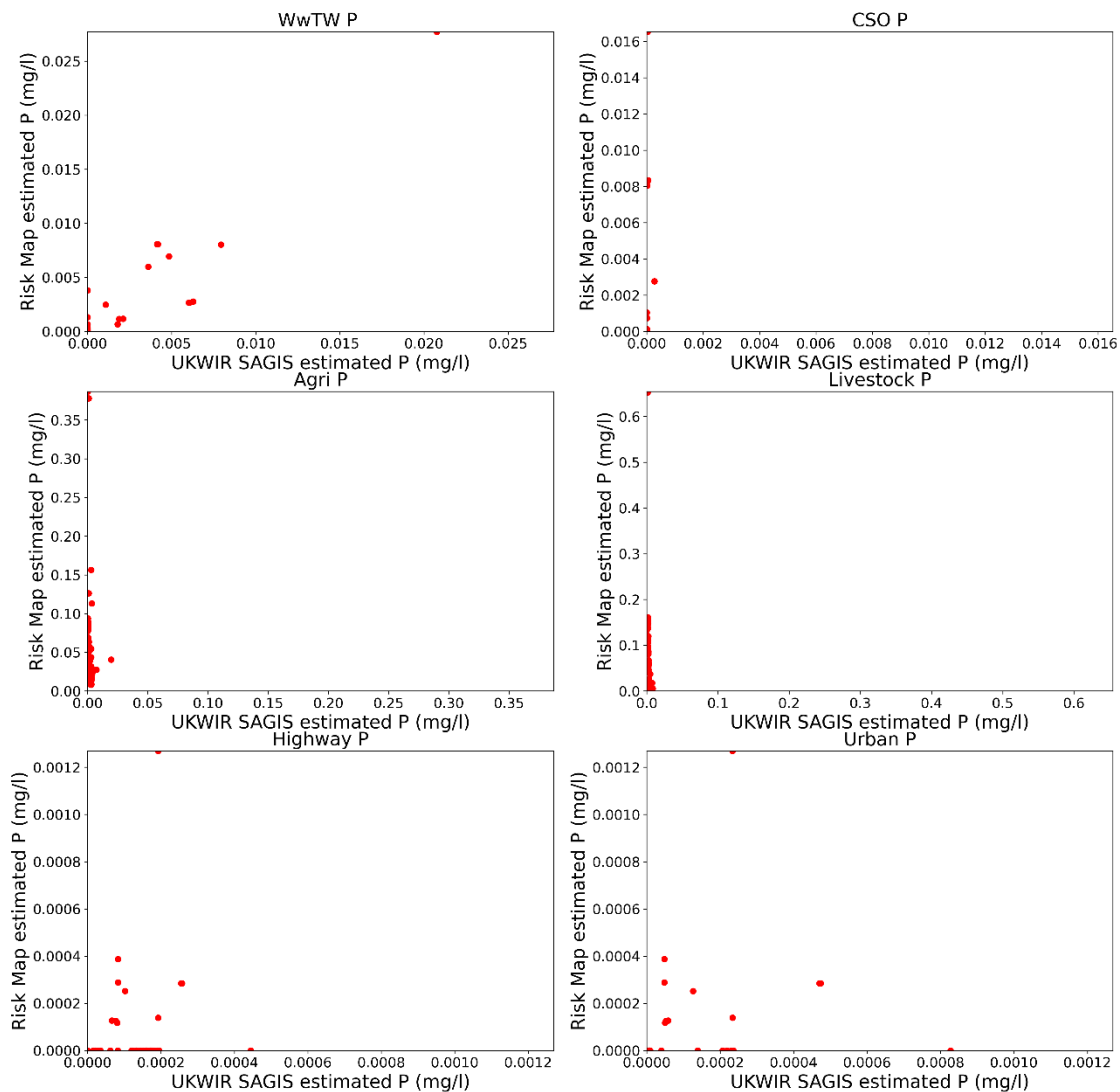
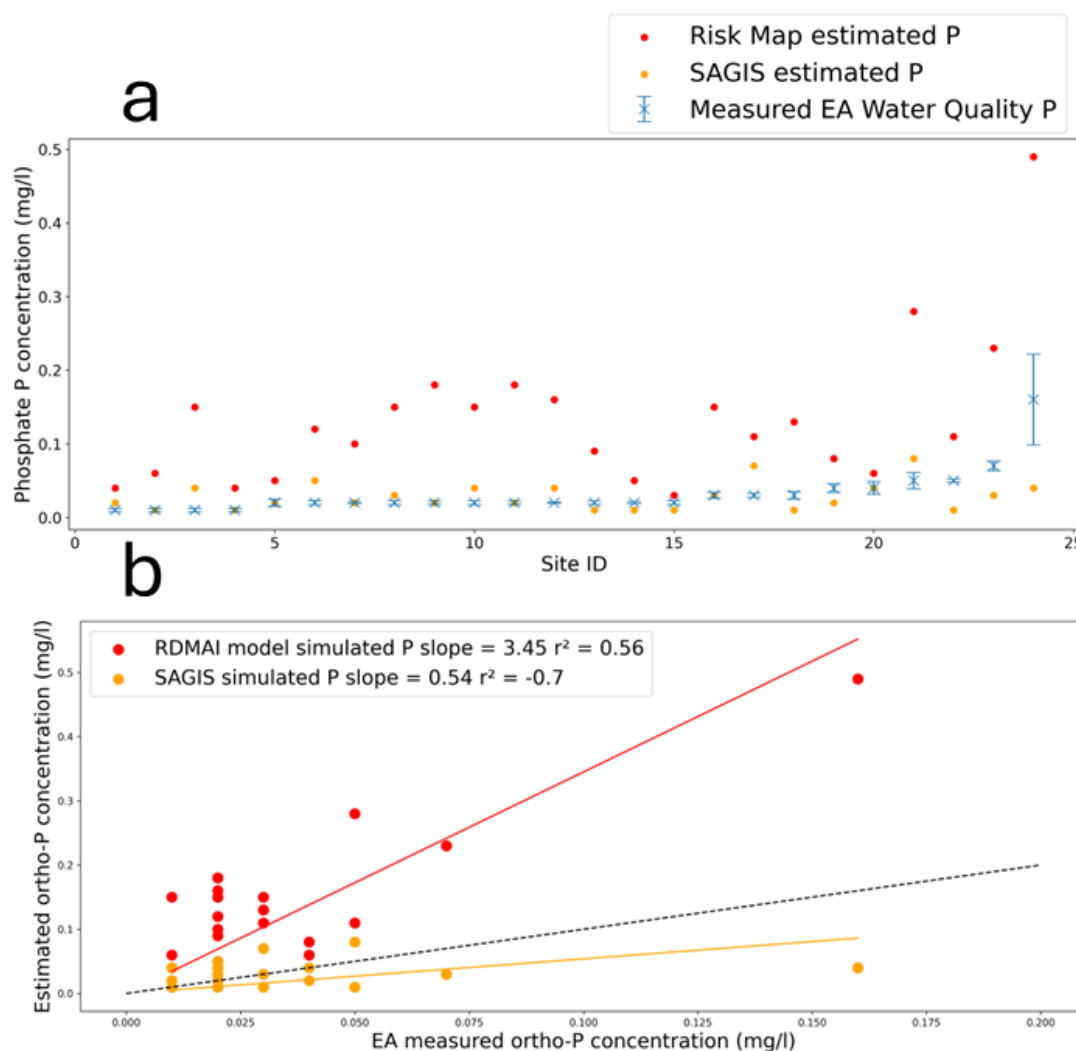




Figure 2.14 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Upper Eden and Eamont - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

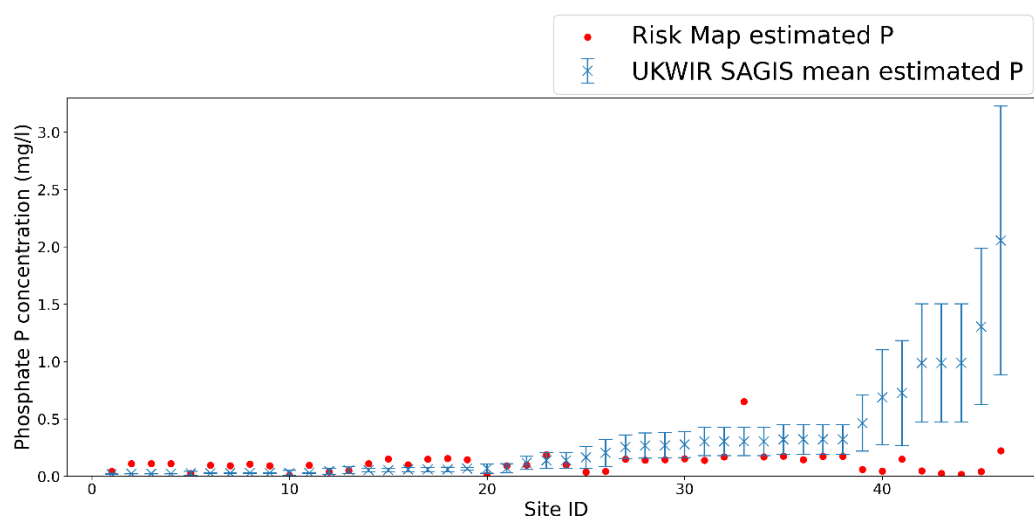


2.4.6 Upper Don

The Upper Don has a relatively high rainfall and is also relatively urbanised.



Figure 2.15 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Upper Don. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P

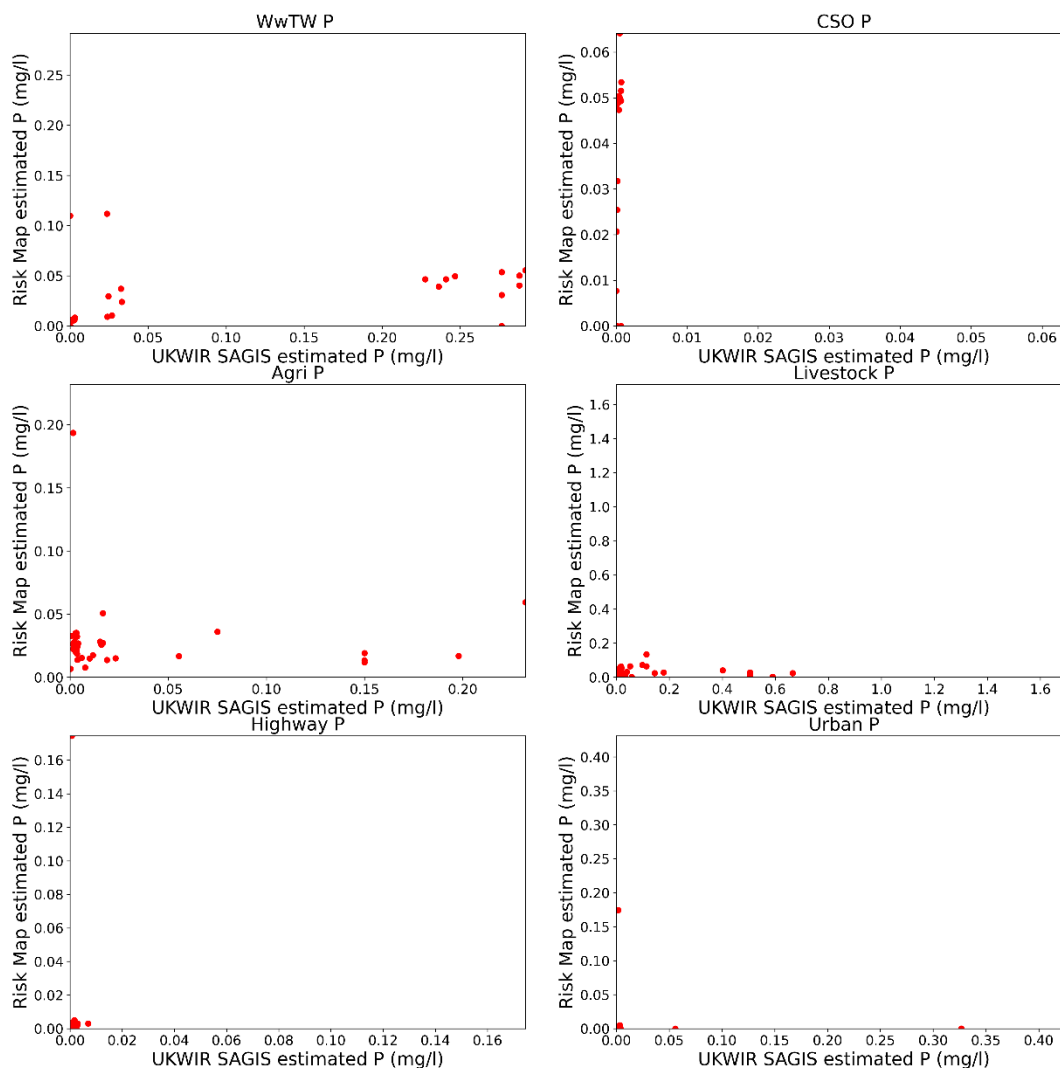


As shown in Figure 2.15, the RDMAI Risk Map tended to underestimate orthophosphate-P for intermediate and higher orthophosphate-P estimates by the UKWIR SAGIS model. When looking at the reasons for differences in Figure 2.16, we can see that:

- The estimates of WwTW contributions to orthophosphate-P concentration by the UKWIR SAGIS model exceed that by the RDMAI Risk Map.
- The RDMAI Risk Map tends to overestimate the orthophosphate-P concentrations of CSOs relative to that by the UKWIR SAGIS model.
- The UKWIR SAGIS model estimates of crop agriculture and livestock orthophosphate-P concentrations exceed those of the RDMAI Risk Map.
- Estimate of highway and urban runoff concentrations of orthophosphate-P by both models were relatively small and consistent.

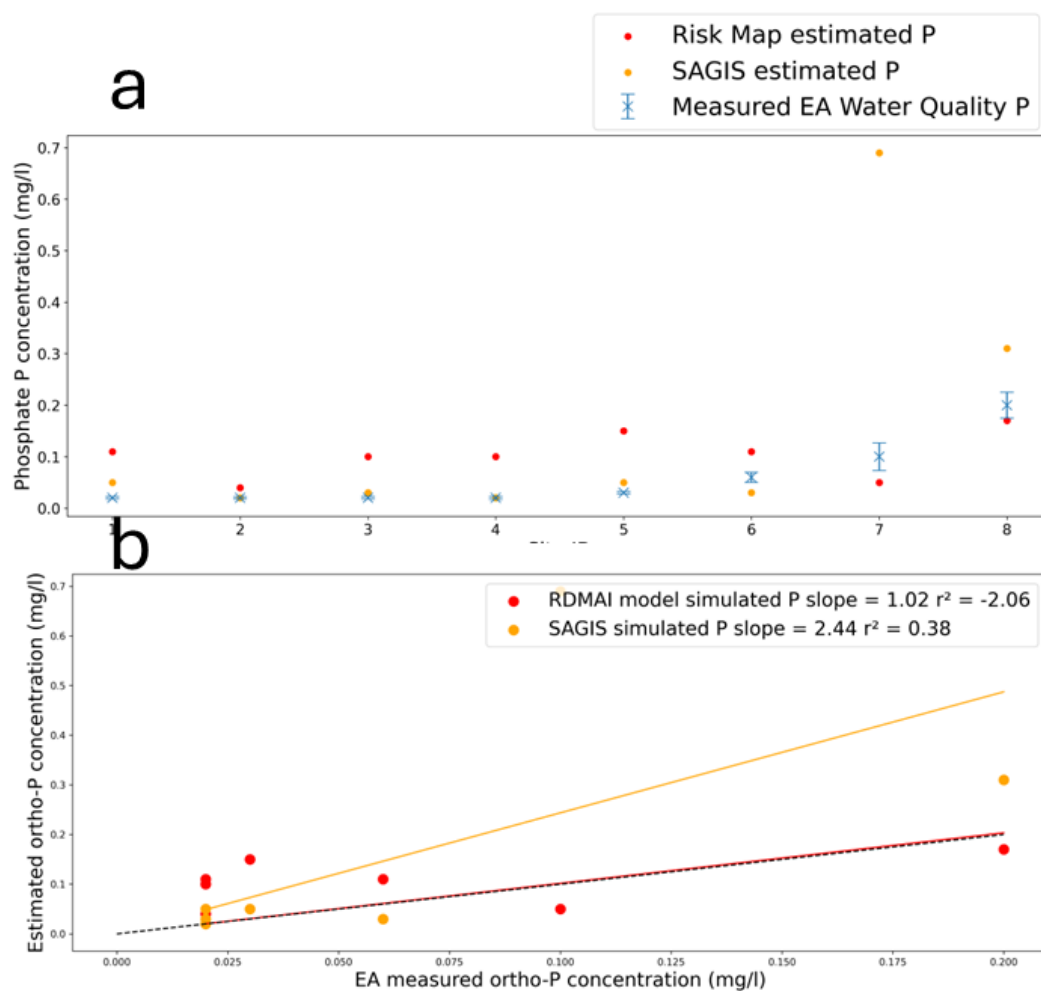


Figure 2.16 RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Upper Don



As shown in Figure 2.17 (a), both models have orthophosphate-P estimations that were similar to the EA measured means for the limited sites evaluated in the Upper Don, although there was one site (site 7) in which the estimate by the UKWIR SAGIS model far overestimated orthophosphate-P. As shown in Figure 2.17 (b), both models showed relatively weak linear relationships to measured orthophosphate-P, although the UKWIR SAGIS model tended to overestimate overall.

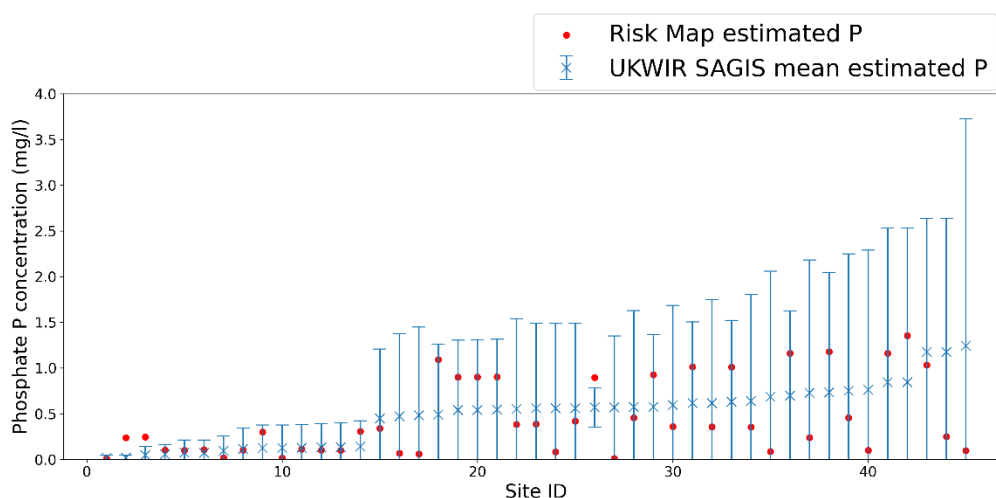
Figure 2.17 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Upper Don - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.



2.4.7 Erewash River

The Erewash River catchment is urbanised, with a relatively lower rainfall.

Figure 2.18 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Erewash River. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



As shown in Figure 2.18 estimates of orthophosphate-P concentration by the RDMAI Risk Map for the Erewash River were consistent with those by the UKWIR SAGIS model for most sites. However, it should also be noted that the confidence intervals for SAGIS in this catchment are relatively wide in comparison to (for instance) Eamont in the North West. When looking at individual contributions in Figure 2.19, it is clear that:

- Both models produced relatively consistent concentrations of orthophosphate-P from WwTWs, while the RDMAI Risk Map produced higher CSO concentrations.
- The RDMAI Risk Map tended to produce lower contributions of crop agriculture and livestock orthophosphate-P concentrations.
- The RDMAI Risk Map tended to produce lower concentrations of highway and urban orthophosphate-P.

As shown in Figure 2.20 (a), orthophosphate-P estimates of both models were relatively similar to EA monitored orthophosphate-P for most sites. However, the RDMAI Risk Map showed a stronger linear relationship to measured orthophosphate-P (Figure 2.20 (b)).



Figure 2.19 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Erewash River

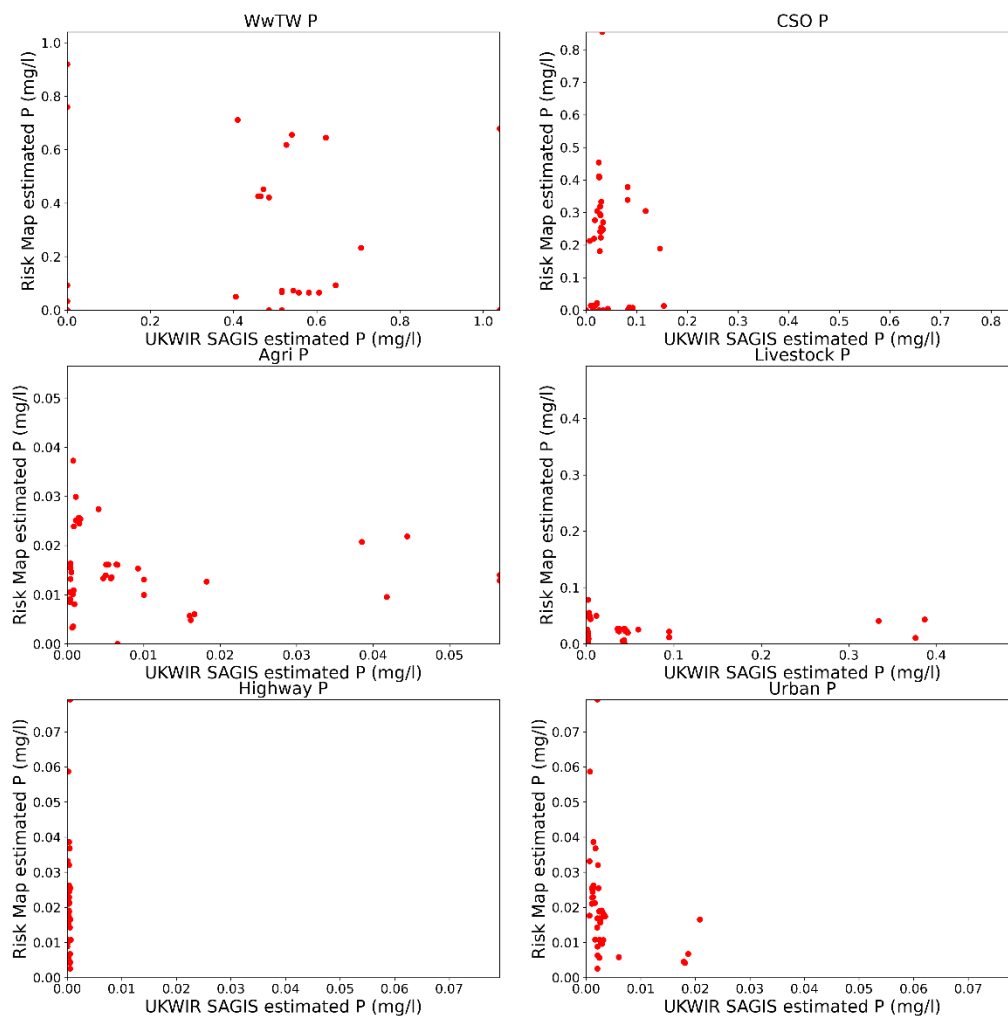
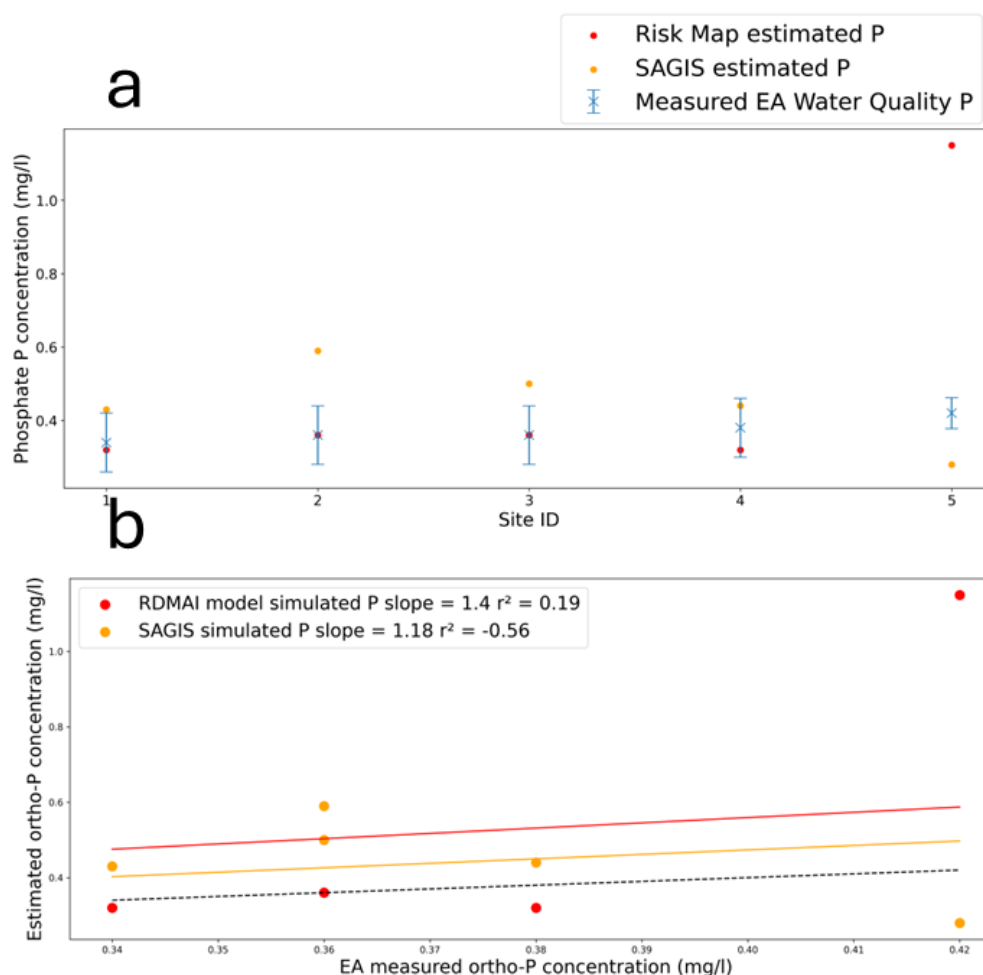


Figure 2.20 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Erewash River - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

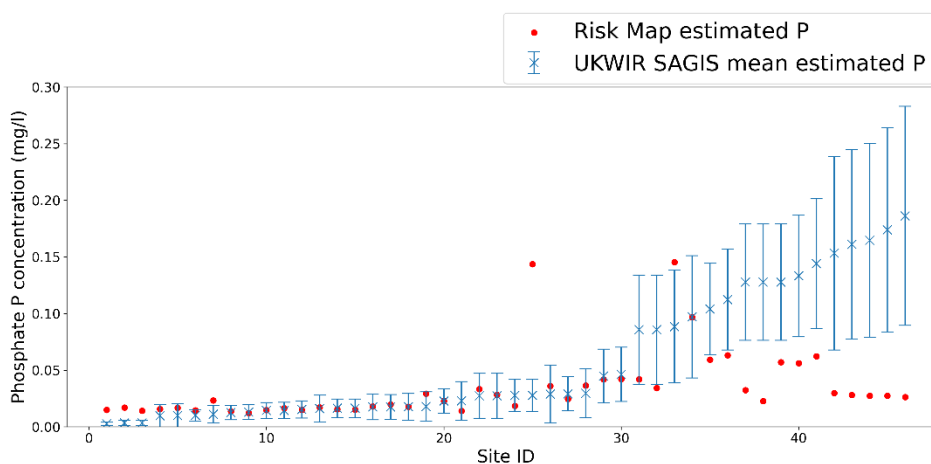


2.4.8 Sections of River Wye

These are relatively rural catchments with moderate rainfall.



Figure 2.21 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for sections of the River Wye. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



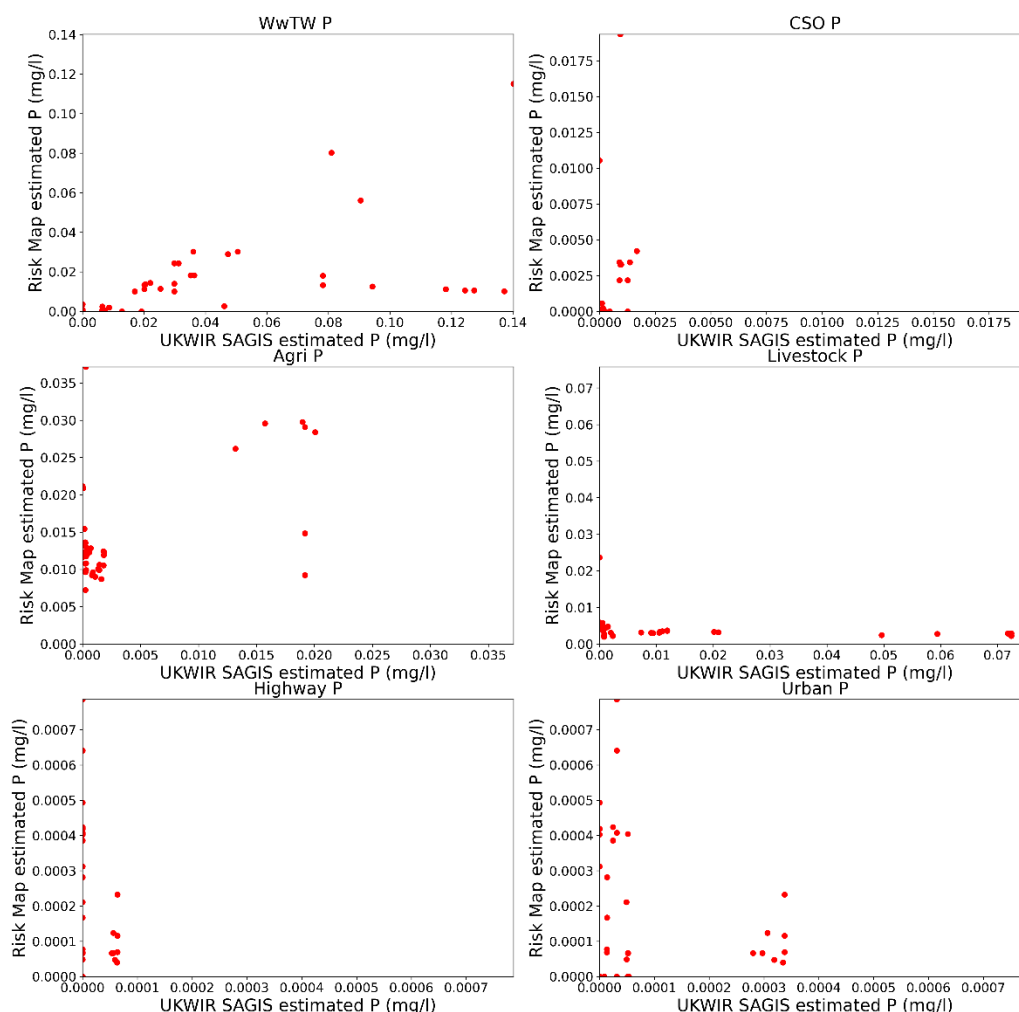
As shown in Figure 2.21, the RDMAI Risk Map estimates of orthophosphate-P tended to be below those of the UKWIR SAGIS model for sites with higher estimates. When looking at reasons for differences in Figure 2.22, it is clear that:

- The RDMAI Risk Map estimates of WwTW and CSO concentrations of orthophosphate-P were underestimated and overestimated relative to those by the UKWIR SAGIS model, respectively.
- Estimations of crop agriculture concentrations of orthophosphate-P were consistent between the two models, whereas the RDMAI Risk Map estimates lower the livestock concentrations.
- Both models gave relatively minor concentrations of highway and urban runoff orthophosphate-P that were relatively consistent.

Unfortunately, no EA monitoring sites could be identified for this benchmark catchment, so a comparison of observed orthophosphate-P could not be performed.



Figure 2.22 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for sections of the River Wye



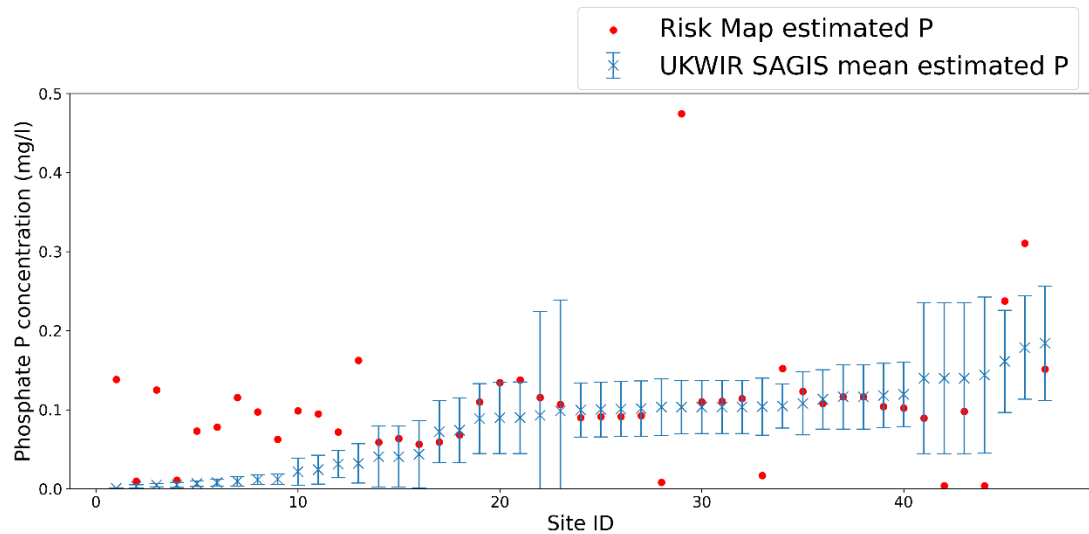
2.4.9 Wensum

The Wensum catchment has some urban area but is mostly rural and has a relatively low rainfall.

As shown in Figure 2.23, RDMAI Risk Map estimates of orthophosphate-P were consistent with those of the UKWIR SAGIS model for sites with moderate and high concentrations, while the RDMAI Risk Map tended to overestimate orthophosphate-P for sites with lower concentrations.



Figure 2.23 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Wensum. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P

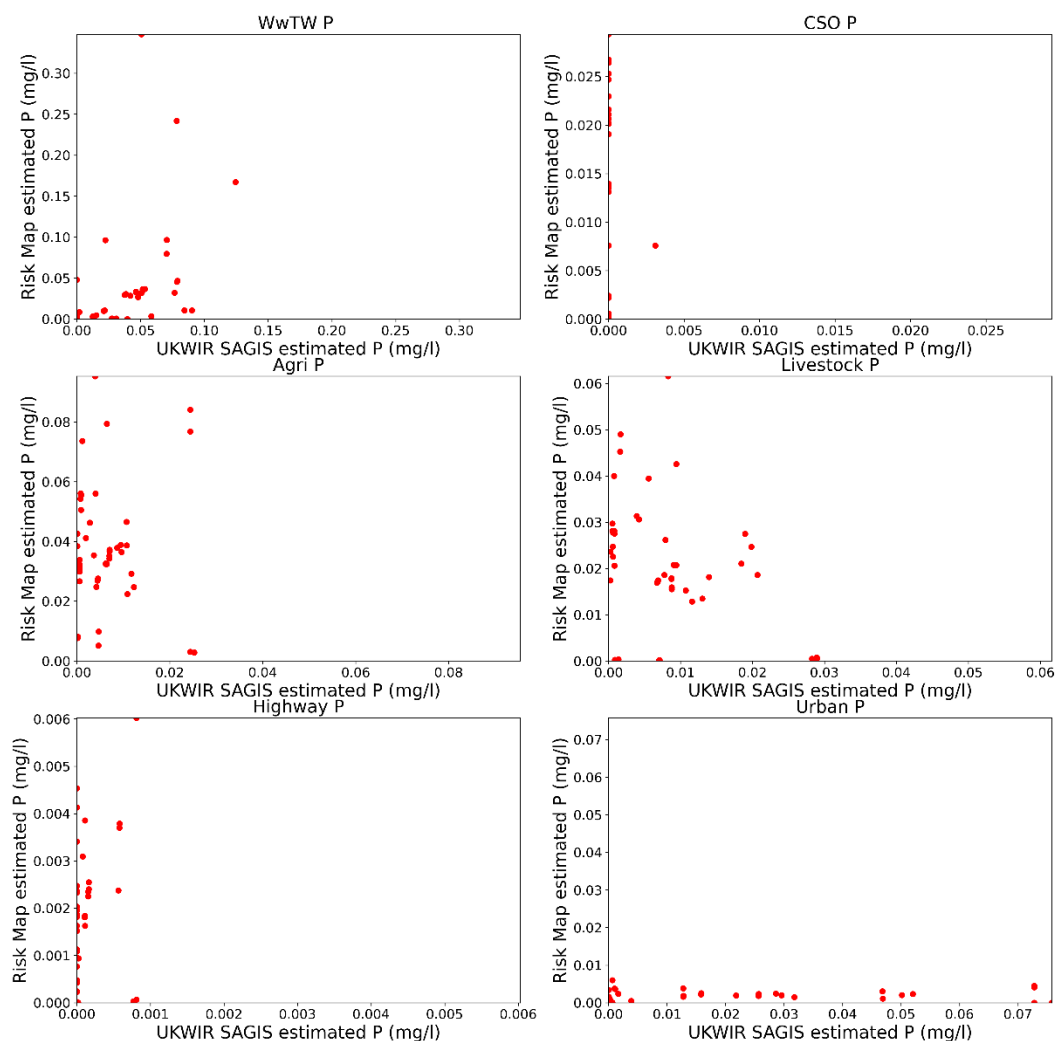


When looking for reasons for differences in Figure 2.24, it is clear that:

- Estimates of WwTW orthophosphate-P concentrations by the two models are consistent, whereas the RDMAI Risk Map tends to estimate higher CSO concentrations.
- The RDMAI Risk Map tends to estimate higher concentrations of crop agriculture and livestock orthophosphate-P.
- The RDMAI Risk Map tended to overestimate and underestimate highway and urban runoff concentrations of orthophosphate-P relative to the UKWIR SAGIS model.

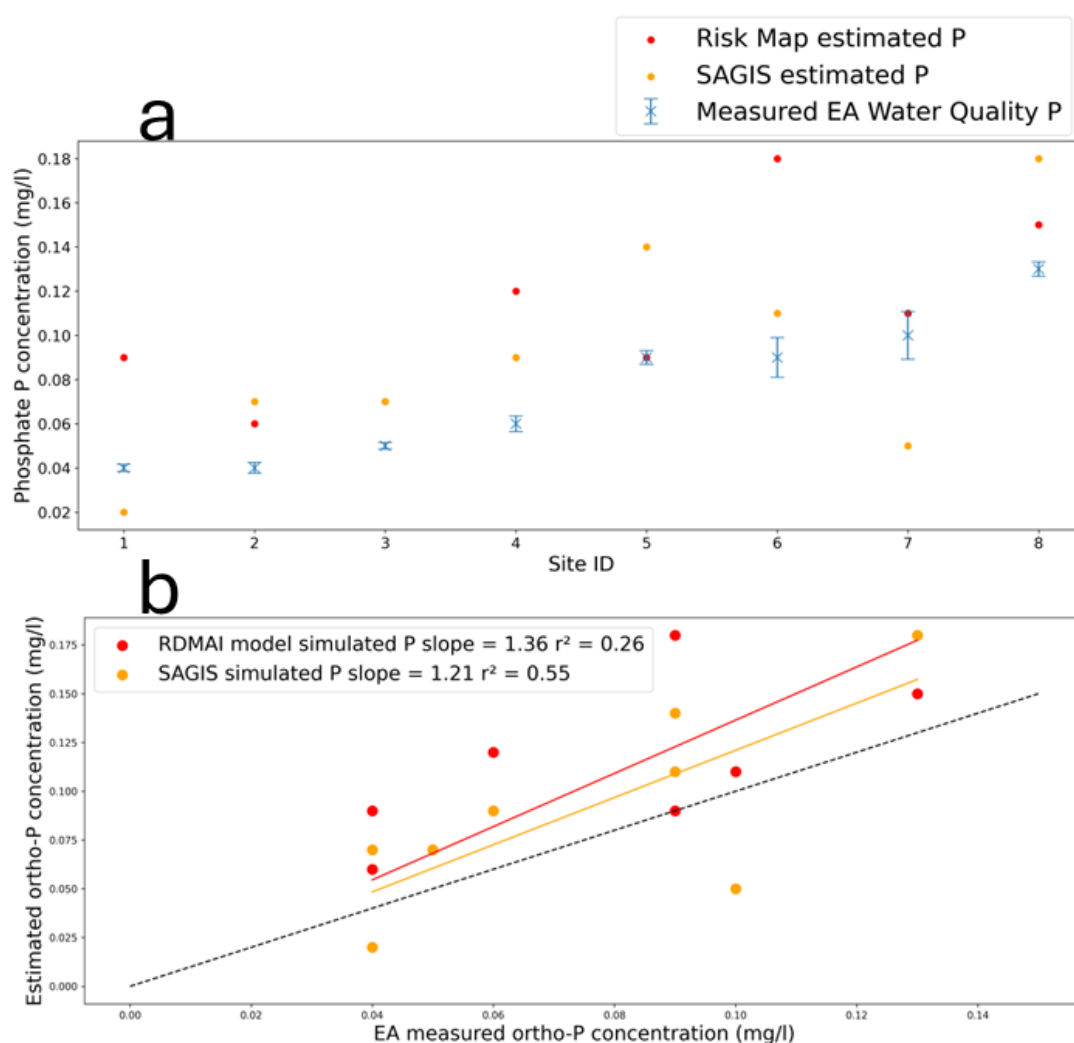


Figure 2.24 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Wensum



As shown in Figure 2.25 (a), both models slightly overestimate orthophosphate-P relative to the EA monitored orthophosphate-P. As shown in Figure 2.25 (a), both models show a linear relationship to measured orthophosphate-P.

Figure 2.25 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Wensum - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.



2.4.10 Ouse

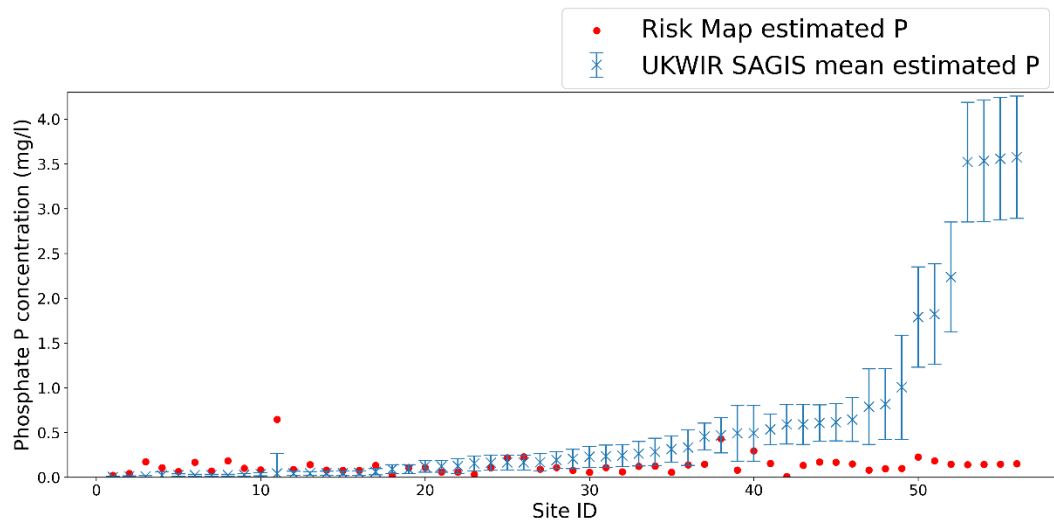
The Ouse in Southeast England has some urban area and a relatively high rainfall.

As shown in Figure 2.26, the RDMAI Risk Map tended to far underestimate orthophosphate-P in the Ouse relative to the UKWIR SAGIS model. When looking at reasons for differences in Figure 2.27, we can see that:

- The RDMAI Risk Map far underestimated WwTW contributions to orthophosphate-P concentration relative to the UKWIR SAGIS model.



Figure 2.26 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Ouse. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



- The estimates of CSO concentrations of orthophosphate-P by the RDMAI Risk Map exceeded that by the UKWIR SAGIS model.
- The RDMAI Risk Map tended to underestimate the concentrations of agricultural orthophosphate-P relative to the UKWIR SAGIS model.
- The RDMAI Risk Map over- and under-estimated the orthophosphate-P concentrations of highway and urban runoff orthophosphate-P relative to the UKWIR SAGIS model, respectively.

As shown in Figure 2.28 (a), the RDMAI Risk Map tended to slightly underestimate orthophosphate-P relative to the higher EA monitored orthophosphate-P sites, whereas for a couple of sites, the UKWIR SAGIS model far overestimated orthophosphate-P. Figure 2.28 (b) shows that the RDMAI Risk Map has a poor linear relationship with measured orthophosphate-P, whereas the UKWIR SAGIS model showed a stronger relationship.



Figure 2.27 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Ouse

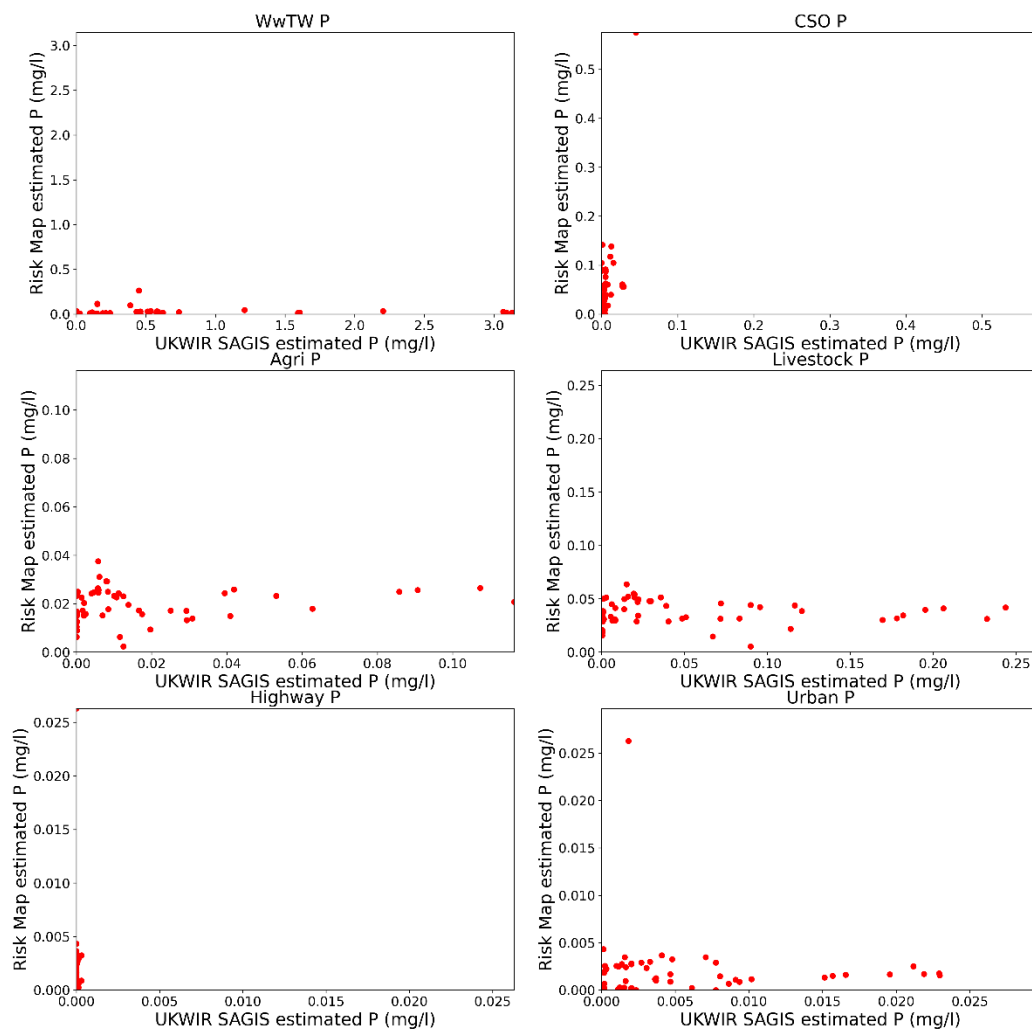
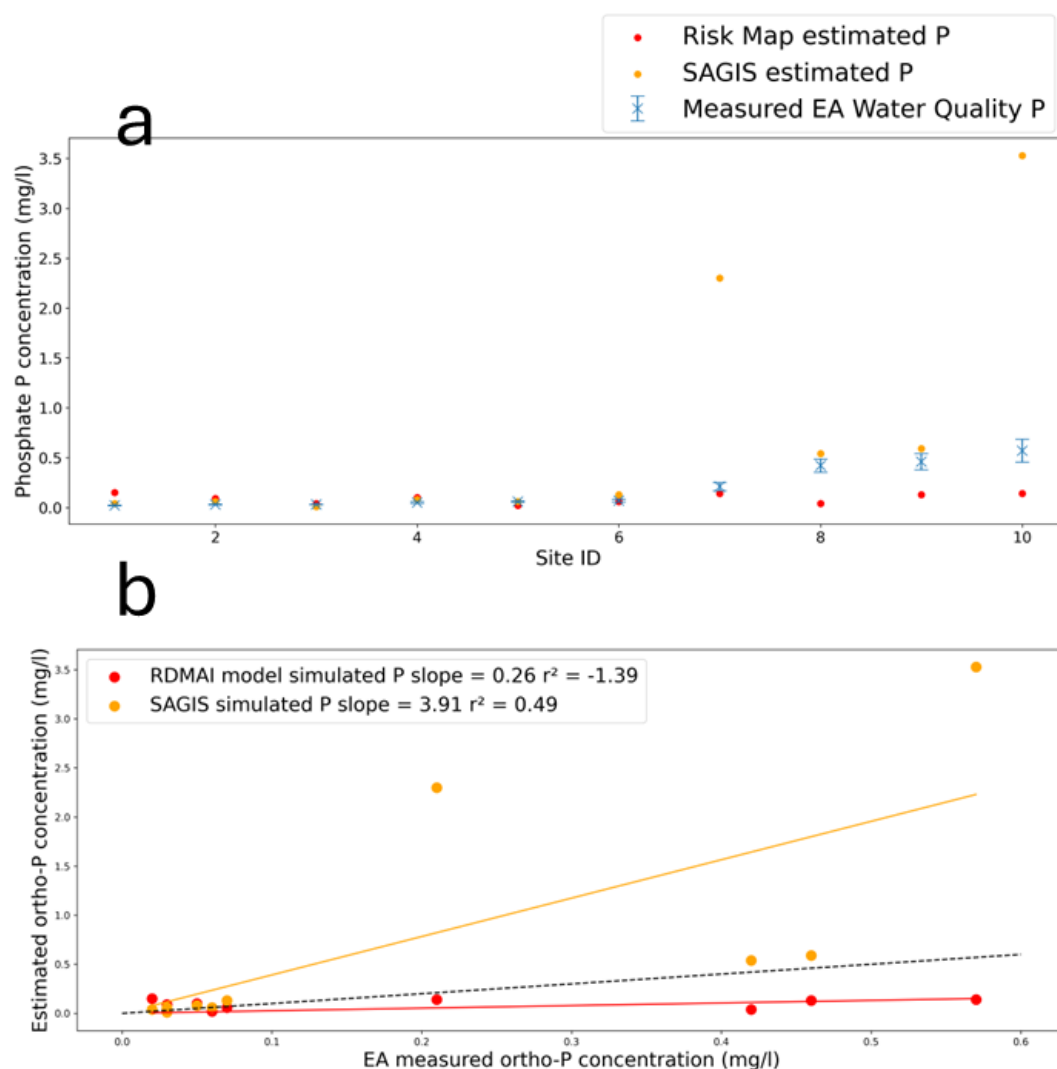




Figure 2.28 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Ouse - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

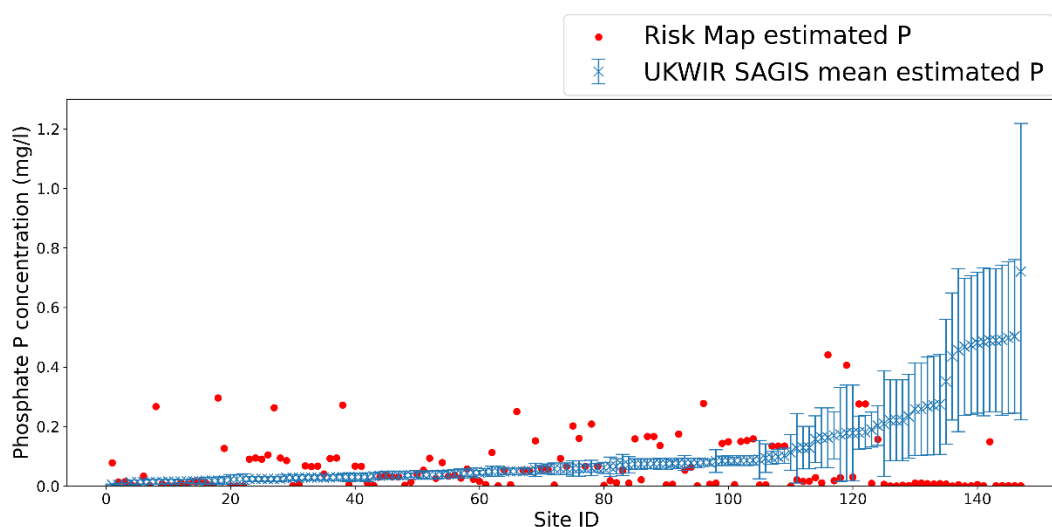


2.4.11 Hampshire Avon

Hampshire Avon is mostly rural with some urban areas. It is also large but has a relatively lower rainfall.



Figure 2.29 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Hampshire Avon. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P



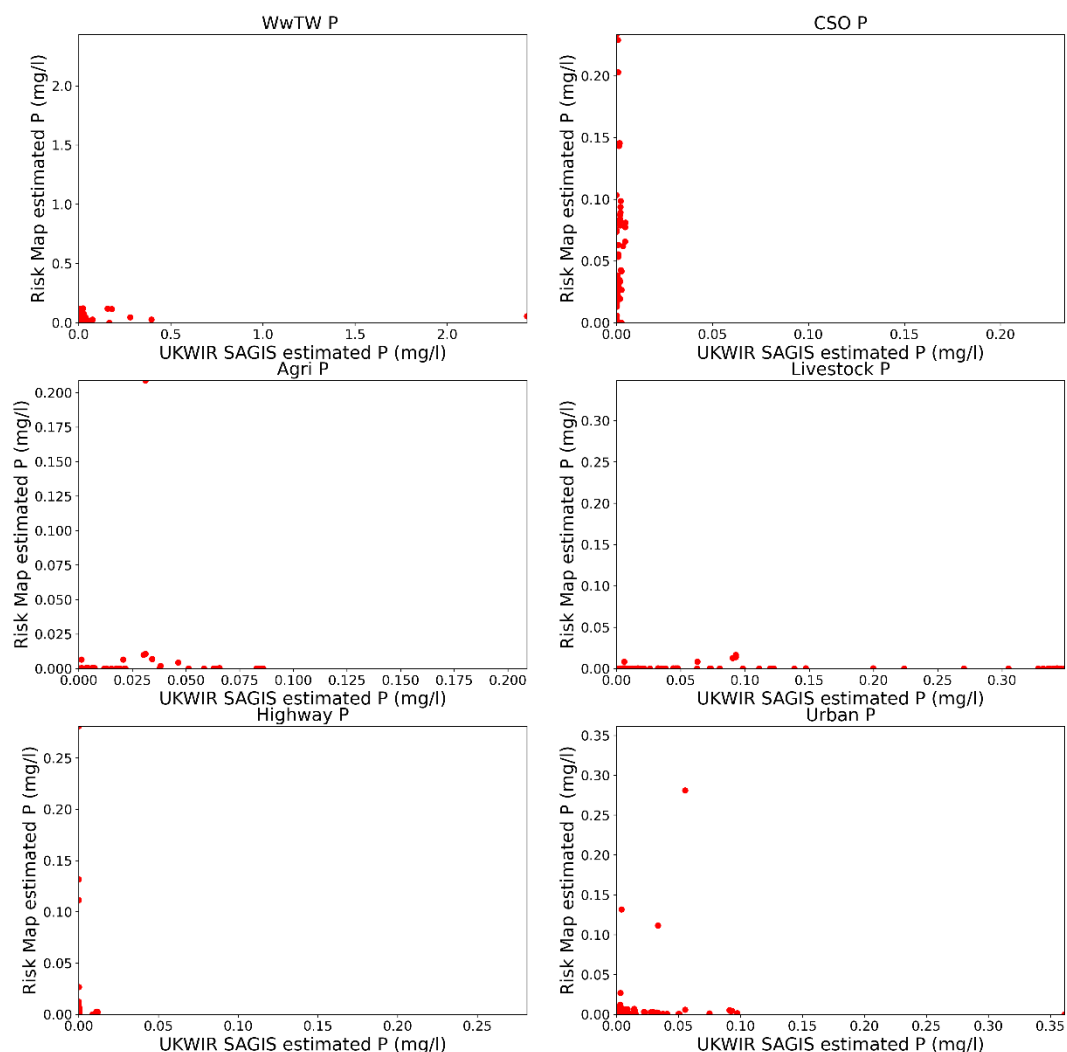
As shown in Figure 2.29, for the Hampshire Avon, there was no obvious consistent pattern between the RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P. For some overall low concentration sites, the two models were consistent, whereas for others, the RDMAI Risk Map tended to overestimate orthophosphate-P. For some sites where the UKWIR SAGIS model estimated high orthophosphate-P, the RDMAI Risk Map generated relatively lower values.

As shown in Figure 2.30, it is clear that:

- Both models generally estimated small WwTW contributions to orthophosphate-P concentration, whereas the RDMAI Risk map generated far larger CSO concentrations.
- The RDMAI Risk Map far underestimated the agricultural concentrations of orthophosphate-P relative to the UKWIR SAGIS model.
- The RDMAI Risk Map generated higher concentrations of both highway and urban runoff orthophosphate-P relative to the UKWIR SAGIS model.



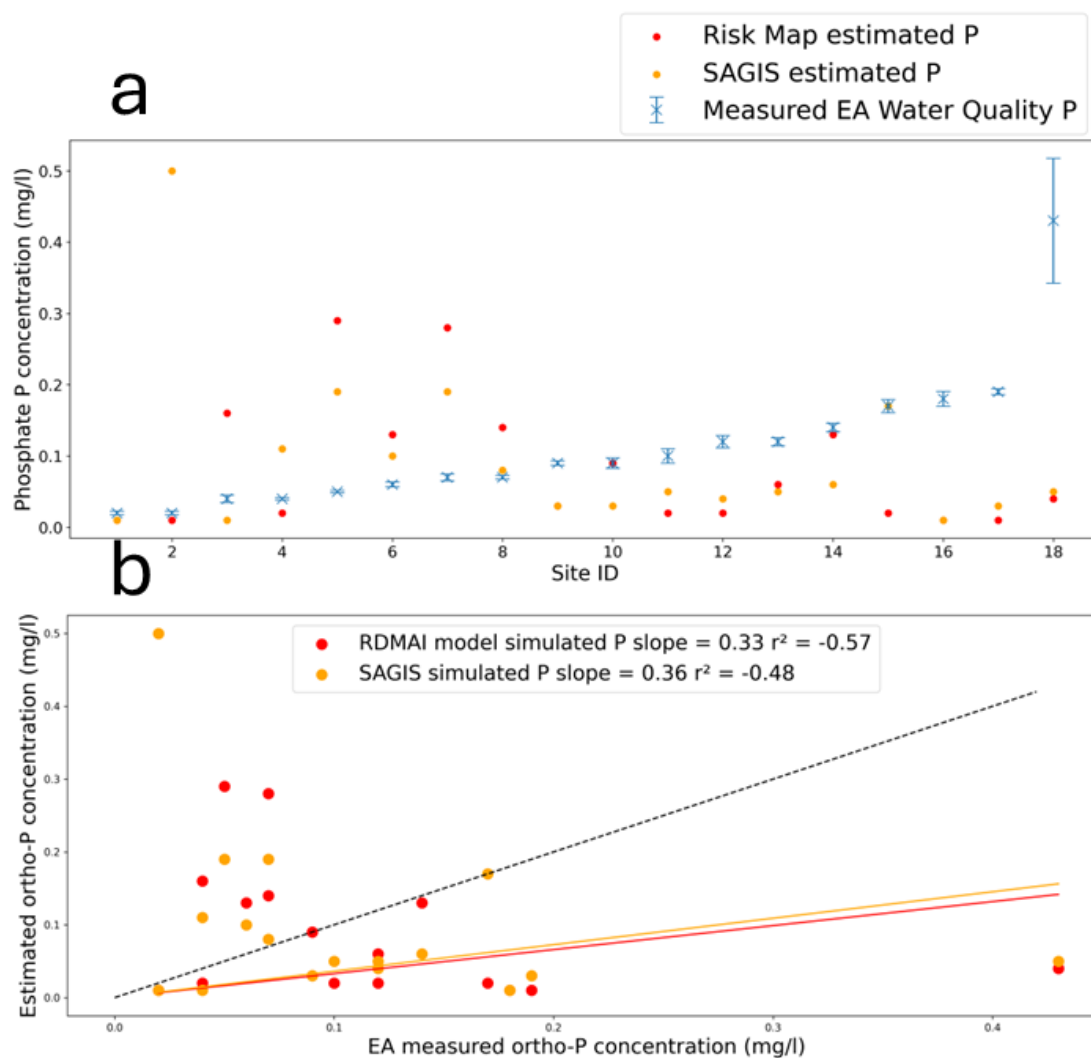
Figure 2.30 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Hampshire Avon



As shown in Figure 2.31 (a), estimates of orthophosphate-P by both models show a mixed performance against the EA observed orthophosphate-P. At sites of higher mean observed orthophosphate-P, both models tended to underestimate, whereas the opposite was true at sites of lower observed orthophosphate-P. This is further illustrated in Figure 2.31 (b), where both models showed poor linear relationships to observed orthophosphate-P.



Figure 2.31 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Hampshire Avon - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.

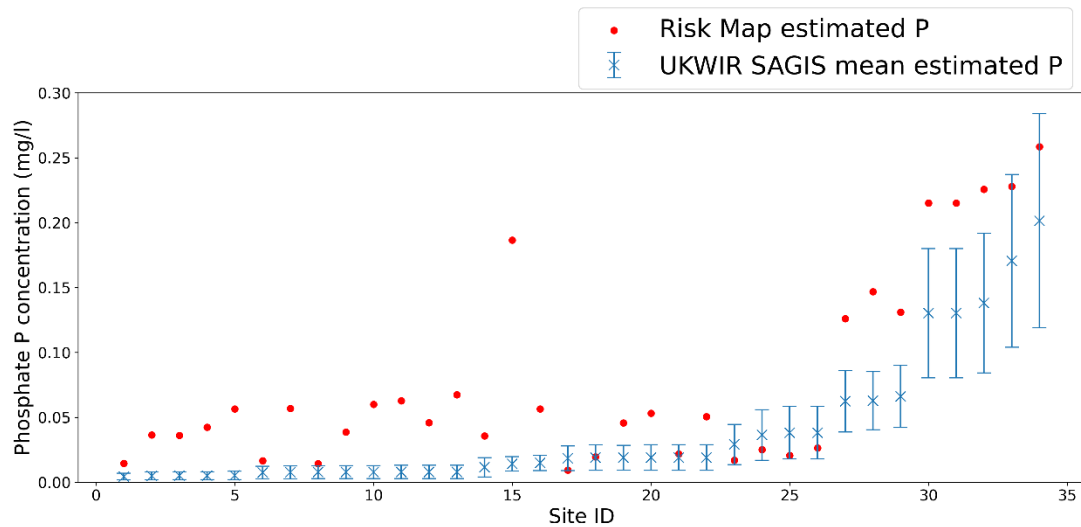


2.4.12 Tavy

Tavy has a high rainfall and is mostly rural.



Figure 2.32 Comparison between UKWIR SAGIS and RDMAI Risk Map mean orthophosphate-P concentration for the Tavy. Sites are ordered by UKWIR SAGIS estimated orthophosphate-P

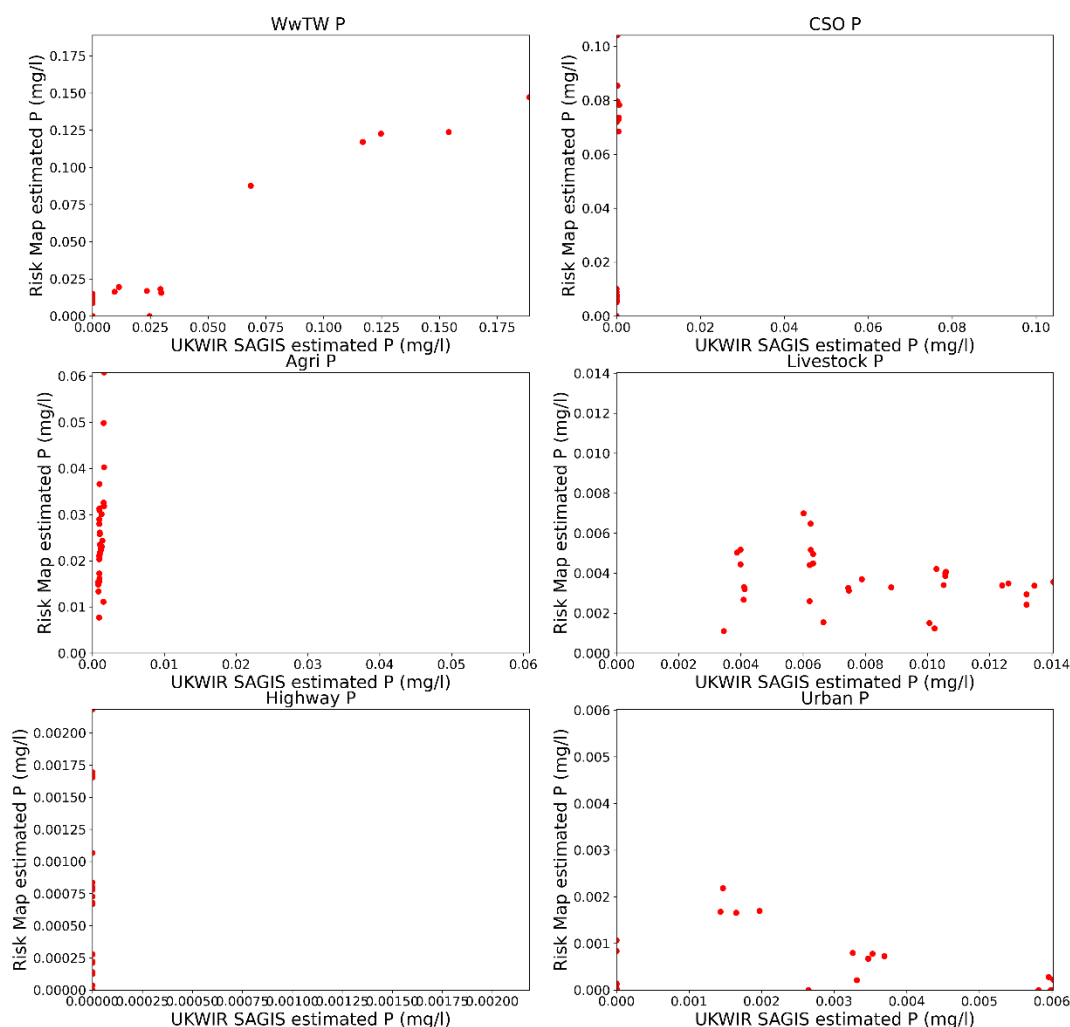


As shown in Figure 2.32, the RDMAI Risk Map tended to overestimate orthophosphate-P relative to the UKWIR SAGIS model. When looking at reasons for differences in Figure 2.33, it is clear that:

- The models showed similar estimates of WwTW concentrations of orthophosphate-P.
- The RDMAI Risk Map far overestimated the orthophosphate-P concentrations of CSOs relative to the UKWIR SAGIS model.
- The RDMAI Risk Map tended to over- and underestimate crop agriculture and livestock concentrations to orthophosphate-P relative to the UKWIR SAGIS model, respectively.
- Both models generated negligible concentrations of highway and urban runoff orthophosphate-P, although the Risk Map estimates of urban runoff orthophosphate-P were lower than those of the UKWIR SAGIS model.

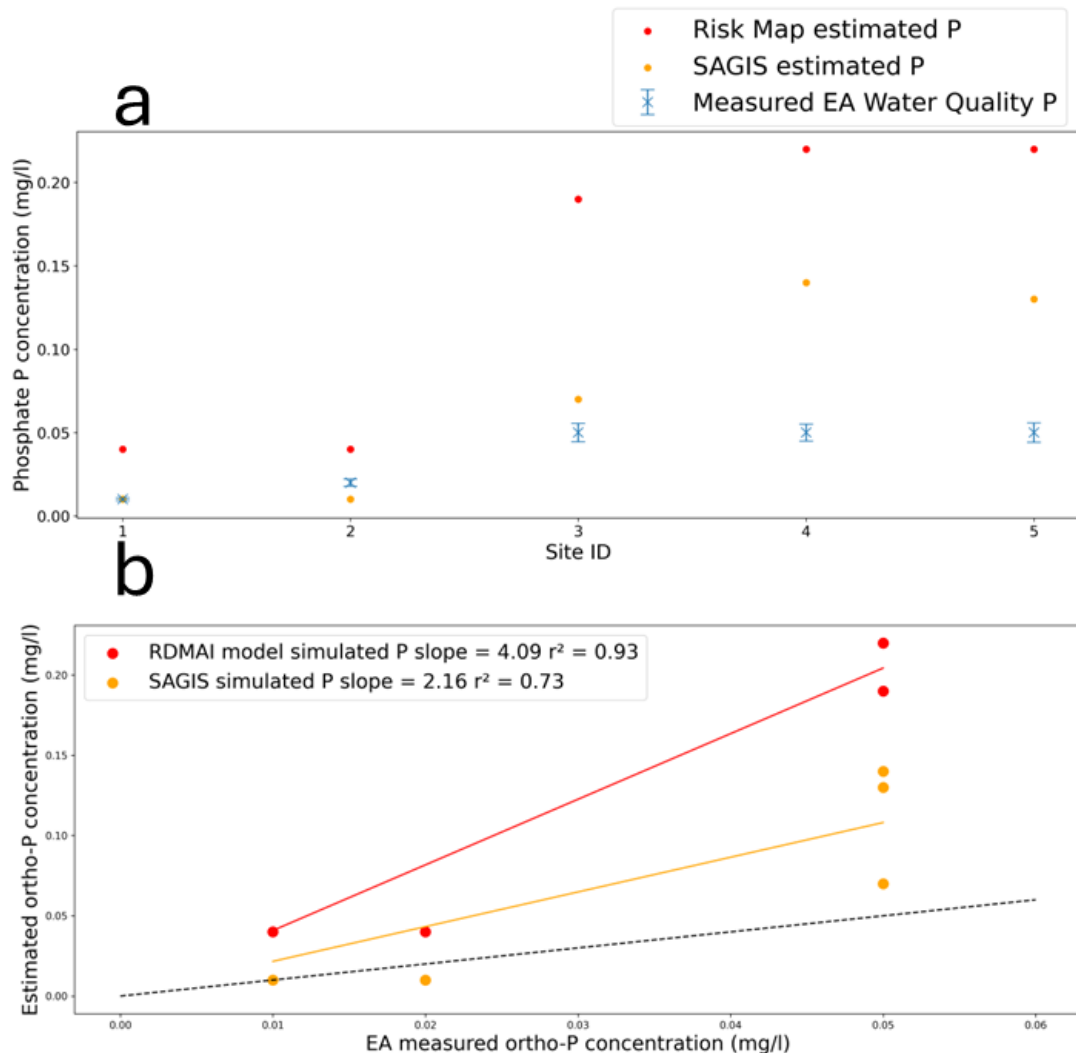


Figure 2.33 X-Y scatter plots for RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P attributable to different pollutant sources for the Tavy



As shown in Figure 2.34 (a), both models tended to overestimate orthophosphate-P relative to EA monitored orthophosphate-P, although the RDMAI Risk Map showed higher overestimation. Both models showed strong linear relationships to measured orthophosphate-P.

Figure 2.34 Comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for the Tavy - a) a per-site comparison with confidence limits around the measured orthophosphate-P; b) as an X-Y plot of estimated verses measured, showing the linear regression relationship.



2.4.13 Comparison of model estimates to EA measured orthophosphate-P for all sites across all benchmark catchments.

Figure 2.35 compares the RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P to EA mean measured orthophosphate-P on a per-site basis for all sites. Estimates of both models appear similar to the measured values at lower orthophosphate concentrations, while at higher concentrations, there is more scatter from both models, although the RDMAI Risk Map estimates look slightly less scattered. This is reflected in the X-Y plot in Figure 2.36, which shows the RDMAI Risk Map to have a stronger linear relationship to measured orthophosphate-P. However, the gradient of the linear regression applied to the UKWIR SAGIS model is closer

to the 1:1 line, whereas that of the RDMAI Open Risk Map indicates an overall over-estimation of orthophosphate-P.

Figure 2.35 Per-site comparison of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P with Environment Agency monitoring point measures of orthophosphate-P for all benchmark catchments. Sites are ordered by increasing measured orthophosphate-P

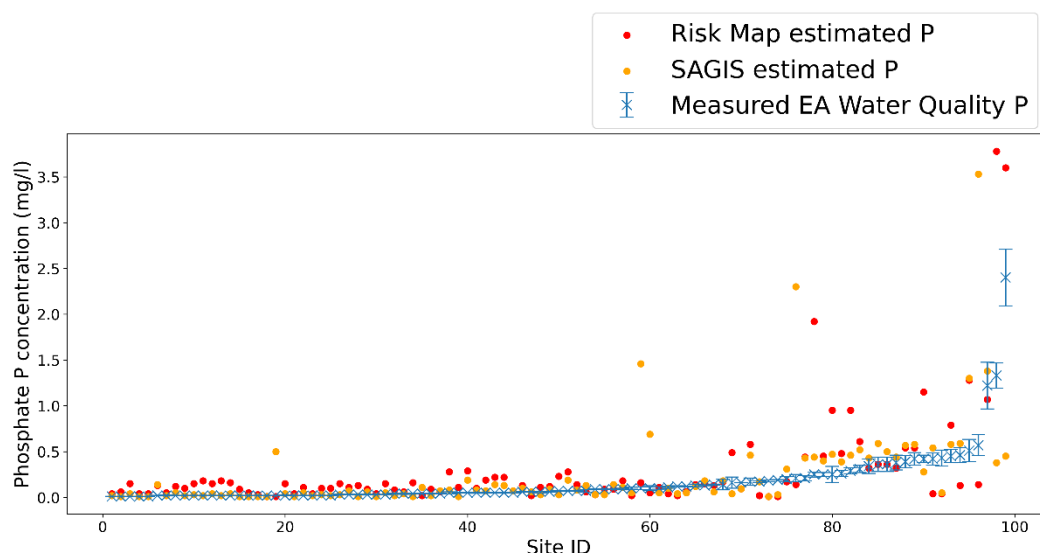
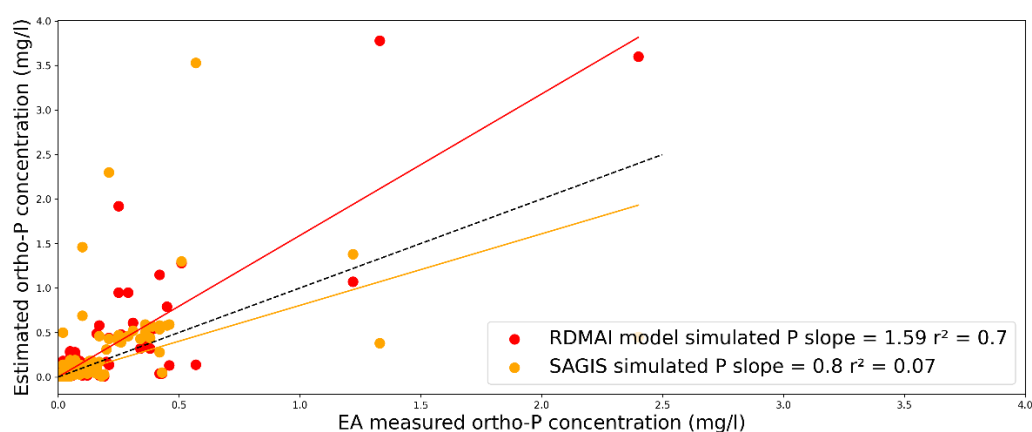


Figure 2.36 X-Y scatter plot of RDMAI Risk Map and UKWIR SAGIS estimates of orthophosphate-P versus Environment Agency monitoring point measures of orthophosphate-P for all benchmark catchments.



2.4.14 Summary of differences between the two models

Table 2.3 shows a summary of differences between the two models on a per-benchmark catchment basis. Here, the mean values of the contributions for different pollution sources by estimated by the two models are compared, on a per-catchment basis. These estimates provide



an indication of the order of magnitude difference in phosphate-P contribution by different pollution sources between the two models.

Table 2.3 Summary of differences between the RDMAI Risk Map and UKWIR SAGIS models for the benchmark catchments

Catch.	Annual Rainfall (mm)	Urban Extent (%)	RDMAI Risk Map: UKWIR SAGIS (mean comparisons)					
			WwTW	CSO	Crop	Livestock	Highways	Urban
Ouse Burn	668	58%	N/A	2.5	4.2	1.8	3.7	0.4
Browney	932	7.5%	1.2	19.8	4.7	5.7	N/A	0.2
Skerne	654	20%	0.8	6.0	3.7	2.7	5.0	1.0
Eden Upper & Eamont	1142	1.6%	0.5	163.6	19.4	10.4	8.0	2.1
Upper Don	1131	9%	0.3	117.6	1.0	0.3	7.0	0.2
Erewash River	708	33%	0.6	5.3	1.6	0.4	26.3	5.4
River Wye	706	0.8%	0.4	4.0	4.5	0.3	74.5	2.1
Wensum	672	5.9%	1.0	125.5	6.0	2.4	10.9	0.1
Ouse	835	14.8%	0.1	8.3	1.1	0.7	55.2	0.4
Hampshire Avon	782	4.8%	0.6	40.8	0.2	< 0.0	12.3	0.4
Tavy	1544	3.4%	1.0	238.2	22.6	0.4	N/A	0.2

It is important to stress that these comparisons do not inform which model representation is more accurate. Some common observations that emerged from Table 2.3 are:

- The UKWIR SAGIS model often estimates larger concentrations of WwTW orthophosphate-P than the RDMAI Risk Map.



- The RDMAI Risk Map consistently represents larger concentrations of CSO orthophosphate-P than the UKWIR SAGIS model.
- Differences in agricultural concentrations of orthophosphate-P between the two models appear to be catchment dependent, although overall, the RDMAI Risk Map generates larger concentrations.
- The RDMAI Risk Map produces higher highway concentrations of orthophosphate-P. SAGIS represents highway runoff using the Highways Agency Water Risk Assessment Tool (HAWRAT) and only represents road runoff pollution from highways and A-roads. The RDMAI Open Risk Map represents urban and rural road runoff contributions from trunk roads (roads within 200 m of a river reach). Therefore, these two models represent road/highway contributions of pollutants very differently, which could explain the differences in the benchmarking.
- In terms of urban runoff concentrations of orthophosphate-P, differences between the two models seemed to be catchment dependent, although it seemed the UKWIR SAGIS model often overestimated urban runoff orthophosphate-P relative to the RDMAI Risk Map. The documentation for the UKWIR SAGIS dataset indicates that urban contributions of pollutants are calculated according to rainfall intensity, CORINE land use data, and literature concentrations in urban runoff. Urban concentrations of orthophosphate-P in the RDMAI Open Risk Map are calculated similarly, except the Ordnance Survey Built-Up Area data are used instead of the CORINE land use data, and the event mean concentrations from the literature used in the RDMAI Open Risk Map might well be different from those used in SAGIS.
- It should be considered that highway and urban run-off is not generally regarded to be a significant contributor to orthophosphate-P. So, this comparison would be arguably more pertinent for benzo(a)pyrene or copper pollutants.

2.4.15 Summary of model differences and performance in relation to observed orthophosphate-P

To compare the RDMAI Risk Map and UKWIR SAGIS model estimates with mean EA observed orthophosphate P, the relative root mean square error (RRMSE) was calculated. This was done for all sites across the different benchmark catchments, because some catchments had too few sites to allow a catchment or regional analysis. Table 2.4 shows a summary of the analysis.



Table 2.4 Comparison of RDMAI Risk Map and UKWIR SAGIS model estimates with EA mean observations for orthophosphate-P for all sites across the benchmark catchments

n	RRMSE(%)	
	RDMAI Open Risk Map	UKWIR SAGIS
99	219%	271%

In Table 2.4, the relative root mean squared error (RRMSE) is calculated as the root mean squared error (RMSE) expressed as a percentage by dividing by the average of the EA observations. As seen in Table 2.4, the RDMAI Open Risk Map and UKWIR SAGIS models overall have a similar performance when benchmarked against EA measured orthophosphate-P in this particular benchmarking exercise.



3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found in Table 1.1, at the start of the report.

3.2 Conclusion

The benchmarking exercise highlighted clear differences in the way the UKWIR SAGIS model and RDMAI Risk Map represent concentrations of orthophosphate-P from different pollution sources. The results clearly show that the UKWIR SAGIS model estimates higher concentrations of WWTW orthophosphate-P, whereas the RDMAI Risk Map estimates higher concentrations of CSO orthophosphate-P. The RDMAI Risk Map also estimates higher concentrations of orthophosphate-P from highways. In terms of concentrations from other sources, consistent patterns between the two models are difficult to establish, with no clear relationship between model performance and simple catchment metrics (rainfall, urbanisation).

In terms of performance of the two models against EA observations of orthophosphate-P, both models perform badly for some benchmark catchments. Overall, the RDMAI Open Risk Map overestimates orthophosphate-P compared to the average EA measures. However, it should be noted that due to the infrequency of EA water quality monitoring, spikes in orthophosphate-P are likely missed, which would act to push the average of monitored values down. This limitation of EA monitoring data was indicated in the previous model validation conducted by RDMAI in a separate report: while the RDMAI Risk Map tended to overestimate the means of EA water quality measures of orthophosphate-P, the model underestimated the means of frequent sonde measures of ammonium-N.

However, according to the benchmarking against EA water quality data here, and considering the benchmarking catchments assessed, the RDMAI Risk Map overall performs slightly better than the UKWIR SAGIS model (RRMSE of 219% for the RDMAI Risk Map versus 271% for the UKWIR SAGIS model).

An identified shortcoming of the RDMAI Risk Map is in its representation of flow; therefore, improved estimates of flow could greatly improve the accuracy of the RDMAI Risk Maps. These can be brought into the Risk Maps by using, for example, the RDMAI Open Flow Model or QUBE estimates. Therefore, the RDMAI Risk Maps show great promise for representing water quality risk in catchments.





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